The Aggregate Impact of Policy Uncertainty in the Green Energy Transition

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October 2026

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Abstract

Federal subsidies for wind and solar are a central instrument of U.S. electricity-sector decarbonization, yet their future availability has historically been uncertain. I study how uncertainty about federal renewable subsidies affects investment and environmental damages in a dynamic spatial general-equilibrium model with stochastic subsidies and irreversible generation capital. I compare transition paths under continuous support and a budget-neutral stochastic subsidy that switches on and off. Relative to the stochastic policy, continuous support produces 13% more renewable capacity and less gas capacity, avoiding $360 billion in climate damages and $46 billion in mortality damages from local air pollution. At the same expected fiscal cost, continuous support generates $292 billion in net benefits. Most of the emissions difference comes from existing coal and gas plants generating more when renewable investment is delayed during subsidy lapses. The effects are largest in regions such as the Midwest, where the subsidy changes whether new renewable capacity is built and existing fossil plants can fill the generation gap during a lapse.

JEL Codes: D81, E22, H23, Q42, Q48, Q54, R13

Keywords: green energy transition, policy uncertainty, irreversible investment, renewable subsidies, spatial general equilibrium

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1 Introduction

The transition from fossil fuels to clean energy represents one of the largest reallocations of capital in history, with global clean-energy investment reaching $2.2 trillion annually (IEA, 2026). For decades, governments have used a range of climate policies to accelerate decarbonization. The majority of the macroeconomics literature that studies the effects of climate policies such as carbon taxes (Golosov et al., 2014), carbon tariffs (Casey et al., 2026a) and renewable subsidies (Arkolakis and Walsh, 2023) takes the future policy regime as given. In practice, developers make long-run investment decisions without knowing whether government support will remain available in the future. The U.S. has subsidized wind through federal tax credits since 1992 and solar since 2006, but that support has repeatedly changed:[1] the wind credit lapsed four times between 1999 and 2015; the 2022 Inflation Reduction Act (IRA) then extended federal clean-electricity tax credits, before the 2025 One Big Beautiful Bill Act (OBBBA) terminated the tax credits for new wind and solar projects. The aggregate cost of renewable policy uncertainty in the United States is unknown.

In this paper, I quantify the macroeconomic and environmental costs of uncertainty about federal renewable subsidies. I build a dynamic spatial general-equilibrium model with irreversible generation capital, convex adjustment costs, and a stochastic renewable subsidy. The model distinguishes generation capacity from electricity generation through merit-order dispatch, allowing changes in subsidy availability to affect emissions through both generation by the existing fossil fleet and investment in new fossil capacity. Electricity trade creates spatial spillovers: renewable investment in one region can displace fossil generation in another. I compute counterfactual transition paths under continuous support and a budget-neutral stochastic subsidy that switches on and off at dates regions do not know in advance. Under the stochastic policy, regions invest in 13% less renewable capacity than under continuous support and build more gas capacity. The additional fossil generation creates $360 billion in climate damages and $46 billion in local air-pollution mortality damages. At the same expected fiscal cost, continuous support generates $292 billion in net benefits.

In the model, an uncertain on-and-off subsidy creates inefficiencies through two mechanisms. First, during a subsidy lapse, a developer can build without the subsidy or wait for support to return. When the return date is uncertain, waiting has an option value: a project built during the lapse does not receive the subsidy, while one that waits can qualify

[1] I refer to the federal production and investment tax credits as renewable subsidies throughout the paper.

when support returns. However, electricity demand continues to grow during policy lapses,[2] so fossil generation increases while renewable investment is delayed. Second, developers can bunch investment into periods in which the subsidy is available. Because convex adjustment costs make it costly to expand capacity rapidly, concentrating investment within eligibility windows creates additional resource costs. Continuous subsidy availability removes the incentives to wait for subsidy renewal or bunch investment around periods when subsidy is available.

Importantly, delaying renewable investment does not require new fossil plants to be built for emissions to rise. Most of the emissions difference in the counterfactuals comes from greater generation by the existing fossil fleet rather than investment in new fossil capacity. While renewable investment is delayed, existing fossil plants generate more: higher utilization of existing coal and gas accounts for 80% of the cumulative emissions difference; the remaining 20% comes from additional gas capacity that would not have been built under continuous support. Renewable investment can resume when the subsidy returns, but the additional gas capacity is irreversible and can continue to generate for decades. The emissions effect of stochastic support therefore depends not only on which technologies are built, but also on how the existing electricity generation capacity is dispatched.

The aggregate results conceal substantial regional heterogeneity. Continuous support has little effect where renewable projects are profitable without federal support or remain unprofitable even with the subsidy. The effects are largest in marginal regions where federal support changes the renewable investment decision and existing fossil capacity can increase generation when investment is delayed. The Midwest combines both features: under continuous support it builds the most additional renewable capacity, while under the stochastic policy its existing coal and gas fleet generates more. I value the local air-pollution consequences of these generation changes using plant-specific marginal damages from AP4. Because coal has substantially higher mortality damages per MWh than gas, the Midwest accounts for more than half of the $46 billion in avoided local air-pollution mortality damages.

Renewable suitability also differs substantially across the U.S. regions. This means that the same renewable capacity can produce different amounts of electricity depending on where it is built. I model electricity trade with transmission constraints. Renewable resources and transmission costs therefore jointly determine where new capacity is built and where its electricity is consumed. Trade also creates spatial spillovers, as renewable investment in one

[2]The U.S. Energy Information Administration projects U.S. electricity demand to grow by roughly 0.9–1.6% per year through 2050, driven in part by data centers and the electrification of transportation and buildings.

region can reduce fossil generation elsewhere. As a result, California, for example, reduces gas generation partly by importing additional renewable electricity from other regions in the Western Interconnection.

I motivate the model with evidence from the history of U.S. renewable tax credits. I show that the production tax credit is large enough to affect whether wind projects are profitable, and that wind investment historically adjusts around changes in subsidy eligibility. For example, following the 2012 expiration deadline, annual additions fell from 13 GW to 1 GW. I also use renewable-equity prices to test whether recent changes in federal support were known in advance. Renewable valuations respond to the IRA agreement, the 2024 election, House passage of the OBBBA repeal, and the Senate's subsequent revision, while fossil-energy equities do not change significantly. I use this suggestive evidence to build a model in which subsidy availability affects renewable investment and investors form expectations over future policy changes without knowing their realized dates in advance.

I calibrate the model using U.S. electricity data from the U.S. Energy Information Administration and the political history of the subsidy. The model has 13 electricity regions that differ in renewable resources, demand, initial generation capacity, and neighbors with whom they can trade electricity. Renewable and gas capital costs, electricity demand, fuel costs, capacity factors, and regional renewable resources are taken from external data and are exogenous in the model; investment-cost parameters are calibrated to national wind and solar capacity growth. The stochastic subsidy process is calibrated separately using observed transitions in unified federal control. The model reproduces untargeted aggregate emissions, wholesale electricity prices, and the generation mix.

I also ask whether continuous support remains valuable when future policy changes are known in advance. In perfect-foresight counterfactuals, firms bunch renewable investment across known eligibility periods, but foresight does not eliminate the value of continuous subsidy. Long subsidy lapses cannot be fully offset by shifting investment across time: renewable investment remains lower during the lapse, while existing fossil plants generate more and additional gas capacity is built. The value of continuous subsidy therefore reflects not only uncertainty about future policy dates, but also the costs of changes in policy availability themselves.

The results are robust to the main uncertainty in the policy calibration. The duration of a future subsidy lapse is not identified by the historical tax-credit record, because past lapses were short and ultimately reversed. I therefore treat the central calibration as a data-

calibrated benchmark and bracket it with alternative forward regimes ranging from a rapidly returning subsidy to permanent repeal. The value of continuous subsidy remains positive throughout, ranging from $56 billion to $385 billion. Excluding local air-pollution damages, the corresponding range is $48 billion to $315 billion. The result also remains positive when climate damages are discounted at the private market rate.

The model abstracts from several margins that could affect the magnitude and composition of the response. Renewable capital costs and coal retirements are exogenous, and the model does not include storage or state clean-energy policies. Section 7.4 discusses these and other omitted mechanisms.

More broadly, the results show that changes in policy can create substantial costs during the green energy transition even when expected government spending is unchanged. When investment is irreversible and costly to shift across time, changes in policy availability affect when firms invest, which technologies are built, and how the existing capital stock is used while investment is delayed. Regional heterogeneity and costly electricity trade create spatial spillovers, so changes in renewable investment in one region can affect production and environmental damages elsewhere. The value of stable climate policy therefore comes not only from reducing uncertainty about future policy, but also from providing continuity during the transition.

1.1 Related literature

This paper contributes to the literature on policy uncertainty and irreversible investment. Uncertainty creates an option to wait and can delay investment (Bernanke, 1983; Dixit and Pindyck, 1994) in the models with sunk costs. Expectations about future policy change can also affect investment before a policy change occurs: Lemoine (2017) shows that anticipated carbon pricing changes current energy investment, while French and Vreugdenhil (2026) show that firms acquire permits before anticipated regulation to preserve future investment opportunities. In electricity markets, Gowrisankaran et al. (2025) study how uncertainty over environmental regulation affects investment in coal generation, and Fried et al. (2022) quantify the macroeconomic effects of uncertainty over whether a carbon tax will be adopted. In my paper, I model both anticipatory and waiting mechanisms. I build on Barradale (2010), who studies uncertainty surrounding repeated expirations of the U.S. production tax credit and argues that uncertainty about renewal can delay wind investment, and Chen (2025), who shows that anticipated expiration accelerates investment before eligibility ends.

My model contains both margins: developers can wait during a subsidy lapse and shift investment toward periods in which support is available. I also separate uncertainty about future policy dates from the effects of on-and-off support itself: even under perfect foresight, known subsidy lapses change investment timing.

A related literature studies the design of clean-energy subsidies. Langer and Lemoine (2022) characterize optimal dynamic adoption subsidies, Aldy et al. (2023) compare investment and production subsidies for wind generation, and Ricks and Kay (2025) study the design of time-limited renewable subsidies. Aldy et al. (2026) examine the role of federal tax policy in U.S. power-sector decarbonization, while Casey et al. (2026b) study the macroeconomic and emissions effects of clean-energy subsidies. I focus on how the stability of federal support over time affects the energy transition and compute the welfare effects in the general equilibrirum.

The paper also contributes to work on the regional effects of policy uncertainty. Poilly and Tripier (2025) show that uncertainty about a common national trade policy has heterogeneous effects across U.S. regions because regions differ in their exposure to trade, while Falk and Shelton (2018) show that policy uncertainty can redirect investment across locations. In my setting, regions face the same federal subsidy process, but its effects are heterogeneous because regions differ in renewable resources and existing generation fleets. Electricity trade also creates spillovers across regions: renewable investment in one region can displace fossil generation in another. Because fossil technologies and local air-pollution damages differ across regions, the environmental value of renewable investment depends partly on where the fossil generation it displaces is located.

Finally, this paper contributes to quantitative general-equilibrium work on climate policy and the energy transition (Golosov et al., 2014; Acemoglu et al., 2012, 2016; Fried, 2018; Arkolakis and Walsh, 2023). Fried (2018) studies how climate policy redirects innovation toward clean technologies, while Acemoglu et al. (2023) show how the availability of natural gas can redirect innovation and alter the long-run energy transition. I take technology costs decline as given and focus on changes in the timing of irreversible generation investment and in the utilization of the existing generation stock: distinguishing generation capacity from electricity generation allows policy to affect environmental damages through both new investment and fossil dispatch.

Roadmap. The rest of the paper is organized as follows. Section 2 describes the institutional background. Section 3 presents empirical evidence on whether subsidy availability affects investment decision and whether the changes in subsidy were foreseen. Section 4 sets up the model and Section 5 calibrates it. Section 6 describes the counterfactuals, Section 7 reports the results, and Section 8 concludes.

2 Institutional background

The U.S. electricity sector has undergone a substantial change in its generation mix over the past two decades and this transition had two margins: switching from coal to natural gas, and then switching from natural gas to renewables. Coal was the leading source of electricity generation in the majority of states in 2001, but by 2025 natural gas had displaced it in most states, while wind had become the leading source in five states (Figure 1). Over the same period, wind and solar installation costs fell substantially, by 55% and 83% since 2010, respectively. Renewable capacity has consequently expanded rapidly, although fossil fuels continue to supply a large share of U.S. electricity.

The transition is geographically uneven because renewable suitability differs across the regions. Figure 2 reports wind and solar capacity factors[3] across the 13 electricity regions used in the model. Wind is especially suitable for the interior regions and Texas, while solar resources are strongest in the Southwest and California. These differences affect the cost of renewable electricity and therefore where wind and solar capacity is built. They also make the effect of a common federal subsidy heterogeneous across space and can create winners and losers.

The Federal Government uses separate tax credits for wind and solar: the Production Tax Credit for wind (a per-MWh payment), and the Investment Tax Credit for solar (a fixed share of capital cost). Both are claimed under a start-of-construction provision that grandfathers the subsidy at the rate that exists when a project breaks ground[4]. The wind PTC lapsed repeatedly after 2000, and each lapse was a brief congressional delay and was renewed retroactively. A wind project that qualifies while the PTC is available retains the

[3]The capacity factor is the ratio of actual electricity generation to the amount a facility would generate if it operated at nameplate capacity continuously.

[4]The wind production tax credit had a placed-in-service rule: a project qualified only if it reached commercial operation before the credit expired, until the American Taxpayer Relief Act (January 2013) replaced it with the start-of-construction rule.

credit over its eligibility window[5] so projects that have already qualified remain insulated from subsequent changes in eligibility. Solar subsidies were stable until 2025. The 2022 Inflation Reduction Act committed both subsidies for a decade. The 2025 One Big Beautiful Bill Act then repealed the subsidies for projects placed in service after 2027, the first full subsidy reversal in their history. (Figure 13) shows the full history of the subsidies.

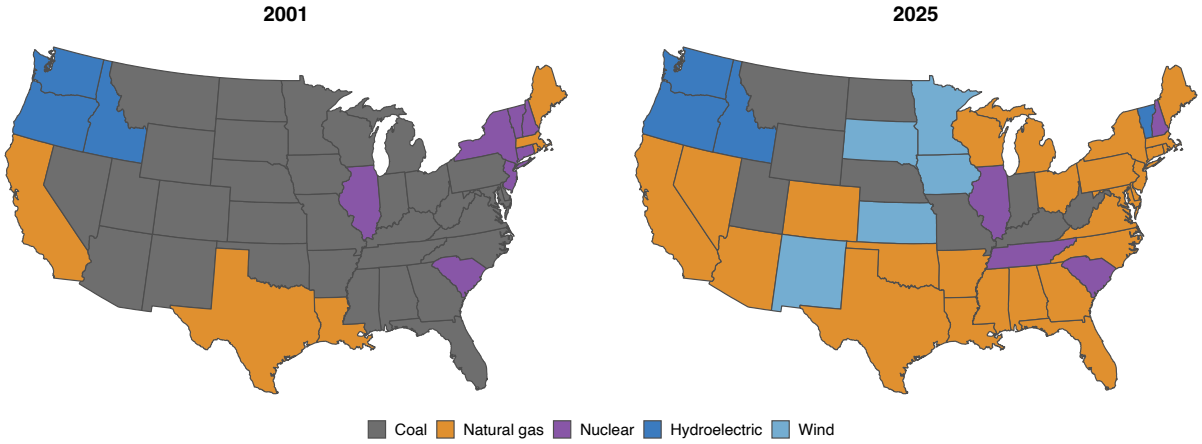


Figure 1: The top source of electricity generation by state, 2001 and 2025.
Notes. Sources: EIA Electric Power Annual, net generation by state and energy source (2001), and EIA-923 (2025).

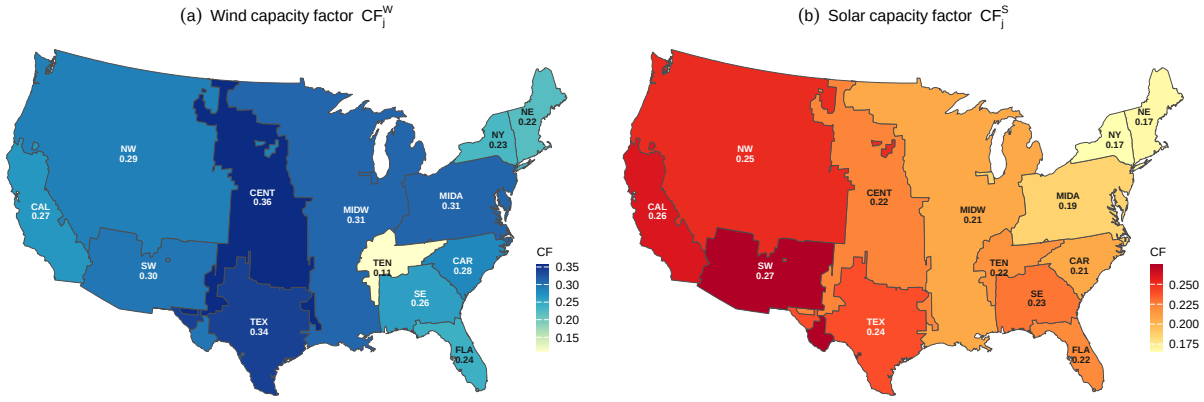


Figure 2: Wind and solar capacity factors across the 13 EIA regions.
Notes. Renewable resource geography across the 13 EIA electricity regions. *Panel (a):* wind (CF_j^W). *Panel (b):* solar (CF_j^S). Source: net generation (EIA-923) over nameplate capacity (EIA-860), 2019–2024.

For most of its history, subsidy availability did not depend on political party in power. Earlier lapses were short legislative interruptions followed by renewal, and the credit sur-

[5]For the PTC, an eligible wind facility generally receives the credit on qualifying generation during its first ten years of operation.

vived periods of unified Republican government in 2003–2006 and 2017–2018. Congressional support subsequently became increasingly polarized: Republican support for legislation containing the credit declined from roughly 75% in 1992 to zero, while Democratic support remained high. After the IRA, the credits became a partisan target, and OBBBA reversed subsidies under unified Republican control. The partisan divide also appears in public opinion: in a 2025 University of Chicago Energy Policy Institute survey, 73% of Democrats but 39% of Republicans described expanding the U.S. clean-energy industry as extremely or very important for the federal government (Energy Policy Institute at the University of Chicago and AP-NORC, 2025).

These changes motivate modeling the relationship between future federal subsidy availability and changes in political control. Section 5 describes how I use the historical political record to calibrate the stochastic policy process.

3 Empirical evidence

I use two facts to justify the model's assumptions about policy uncertainty. First, I show that the federal wind subsidy is large enough to affect project profitability and that wind investment adjusted around changes in subsidy eligibility in the last 25 years. Second, I use financial-market responses to recent federal policy events to test whether investors knew the realized future policy path in advance.

3.1 The subsidy availability affects wind investment

The production tax credit is large relative to wind-project revenue. A wind plant earns the wholesale price of electricity and, during its first ten years of eligibility, the §45 production tax credit of $26/MWh for each megawatt-hour it generates. I combine EIA-860 and EIA-923 data with engineering cost assumptions to calculate project revenue and levelized cost[6] across the observed distribution of wind capacity factors. At the $50/MWh average on-peak wholesale price over 2010–2024 across 8 trading hubs (EIA/ICE), the PTC accounts for 34% of project revenue during the years in which it is paid. The representative project's levelized cost is $53/MWh, slightly above the wholesale price of $50/MWh without the subsidy.

[6]The levelized cost of electricity is the average cost per MWh of generating electricity over a plant's lifetime, including its up-front capital cost and operating costs.

Without the credit, the representative wind project breaks even at a capacity factor of 0.35; the subsidy lowers the breakeven capacity factor to 0.23. Of the 129 GW of operating U.S. wind capacity in the calculation, 48% lies above the unsubsidized breakeven threshold, 35% lies between the subsidized and unsubsidized thresholds, and the remaining 17% lies below the subsidized threshold. Thus, for a substantial share of the observed wind-capacity distribution, the PTC is large enough to move the representative project across the profitability threshold.

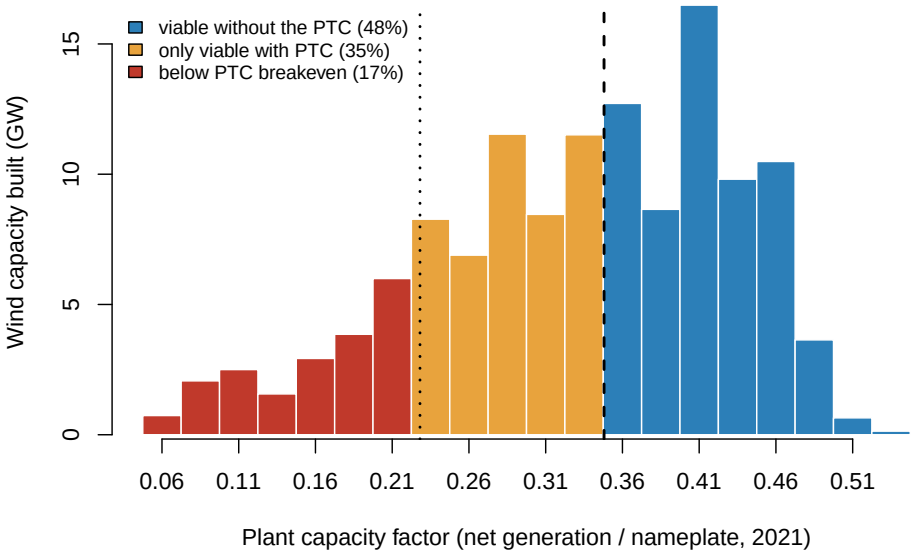


Figure 3: Wind capacity across subsidized and unsubsidized breakeven thresholds.

Notes. Operating U.S. wind capacity (EIA-860, 2021) by realized capacity factor (net generation, EIA-923, over nameplate capacity). The dashed line is the capacity factor at which the representative project breaks even at the $50/MWh wholesale price (0.35); the dotted line is the breakeven capacity factor when the $26/MWh credit is added (0.23). Capacity between the two lines lies between the subsidized and unsubsidized breakeven thresholds under the assumptions of the calculation. Engineering assumptions are reported in Table 11. Sources: EIA-860 and EIA-923.

Wind investment also shows pronounced boom-bust patterns around PTC expiration dates. Annual capacity additions rose before historical deadlines and fell sharply afterward. Following the four unrenewed expiration deadlines in 2000, 2002, 2004, and 2013, annual wind additions fell by 79%, 60%, 71%, and 92%, respectively (Figure 4). The largest decline followed the 2012 deadline, when additions fell from 13 GW in 2012 to 1 GW in 2013. This investment pattern around the PTC has also been documented by Barradale (2010) and Chen (2025).

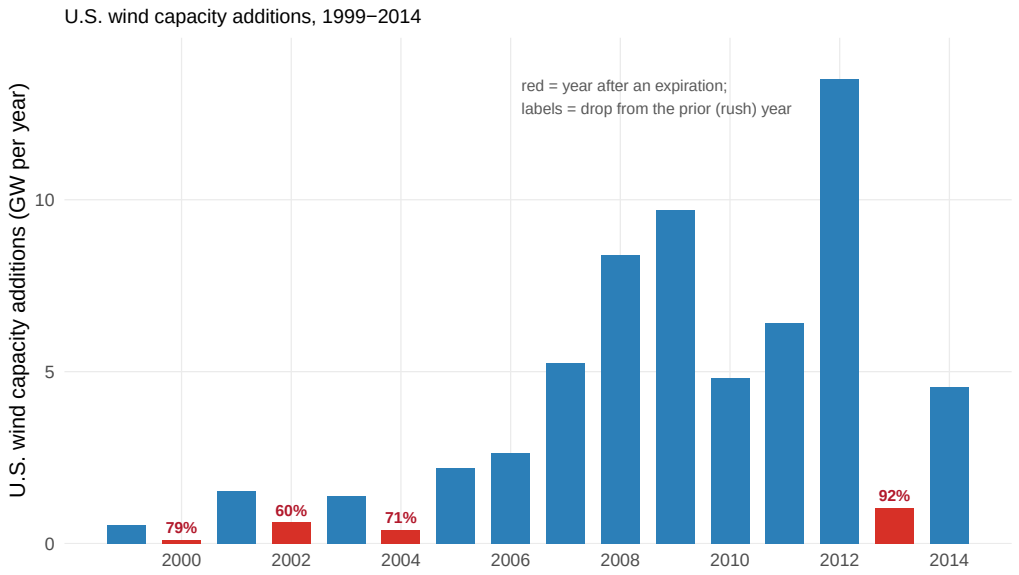


Figure 4: Wind capacity additions around the four PTC expirations.

Notes. National U.S. wind capacity additions, 1999–2014. Bars for the four years following the deadlines (2000, 2002, 2004, and 2013) are shown in red and labeled with the decline from the preceding year. Source: EIA-860.

3.2 Evidence on investors' information about future policy

The second set of evidence explores investors' information about future policy. In the model, investors have rational expectations over future subsidy states and can anticipate the possibility of a policy change, but they do not know the realized future policy path. I use financial-market responses to justify this assumption.

Under full foresight, the realization of the event contains no new information and therefore should not change firm valuations. I consider four events: the Inflation Reduction Act (July 2022), the 2024 election, House passage of the OBBBA repeal (May 2025), and the Senate revision (June 2025). Figure 5 shows the changes for renewable- and fossil-energy equities. I calculate cumulative returns in excess of the S&P 500 from the event day through three trading days after the event. For each event, I summarize the differential response as

$$\Delta_e = R_e^{\text{ren}} - R_e^{\text{fos}}, \tag{3.1}$$

where R_e^{ren} and R_e^{fos} are cumulative abnormal returns for the renewable and fossil portfolios. I also report the same difference over the five trading days before each event, $[-5, -1]$, to show the extent of pre-event movement.

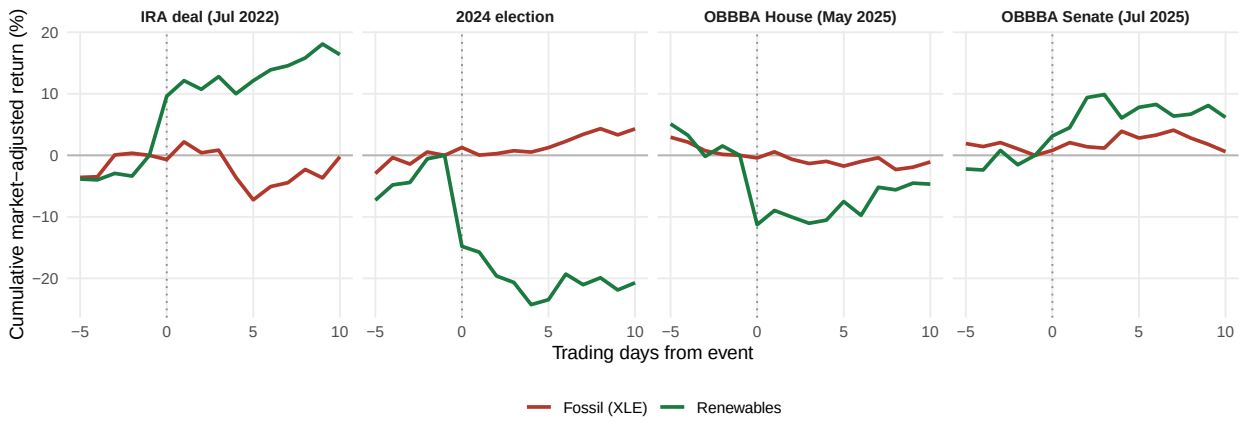


Figure 5: Stock-price reaction to federal policy events.

Notes. Cumulative return in excess of the S&P 500 normalized to zero on the trading day before the event. Source: daily stock prices.

Table 1: Renewable and fossil equity around federal policy events: cumulative abnormal returns.

Event	Renew.	Fossil	Difference $[0,+3]$		Placebo $[-5,-1]$	
	CAR (%)	CAR (%)	$R-F$ (pp)	t	$R-F$ (pp)	t
IRA announcement (Jul 2022)	+12.8	+0.8	+12.0	1.8	+2.2	0.3
2024 election (Nov 2024)	-20.7	+0.7	-21.4	-4.4	+3.9	0.7
OBBBA House (May 2025)	-11.0	-1.3	-9.7	-2.4	-1.7	-0.4
OBBBA Senate (Jul 2025)	+9.9	+1.2	+8.7	1.8	+11.1	2.1

Notes. CAR: cumulative abnormal return relative to the S&P 500. $R-F$ is the difference between the renewable and fossil portfolio CARs, in percentage points. The event window $[0,+3]$ starts from the event day through three trading days after the event. The placebo window $[-5,-1]$ reports the same renewable-minus-fossil difference over the five trading days preceding the event. The renewable portfolio includes TAN, FAN, ICLN, First Solar, Sunrun, Enphase, and NextEra; the fossil portfolio is XLE.

Renewable equities gained 12.8% following the IRA agreement, lost 20.7% after the 2024 election and 11.0% after House passage of the OBBBA repeal, and gained 9.9% when the Senate subsequently softened the repeal (Table 1 and Figure 5). The fossil equities did not change to the same extent. The signs of the renewable-minus-fossil differences track whether an event increased or reduced expected federal support. The pre-event differences are generally smaller, although the Senate event shows substantial movement before the event window, so I do not interpret these estimates as causal event-study effects. I use this exercise as suggestive evidence to motivate the uncertainty in the model: investors can anticipate the possibility of policy changes, but they do not know the realized future subsidy path in advance.

4 Model

I develop a dynamic spatial general-equilibrium model that links stochastic renewable subsidies to investment, electricity generation, and emissions. The economy consists of 13 electricity regions connected through three interconnections. Households own generation capital and make irreversible investment decisions in wind, solar, and gas under uncertainty about future federal subsidies. Final-good firms use capital, labor, and electricity, and production capital is traded nationally. Within each region, installed generation capacity is dispatched in merit order against fluctuating electricity demand, and regions trade electricity within their interconnection. A capacity-adequacy requirement ensures that each region has enough reliable generation capacity to meet forecast demand.

4.1 Environment

Time is discrete and the horizon is infinite, $t = 2010, 2011, \ldots$. There are $J = 13$ regions indexed by j, corresponding to the Energy Information Administration electricity regions and grouped into the Eastern, Western, and Texas interconnections.[7] Each region has a representative household-generator that consumes, supplies labor inelastically, saves in a national capital market, and owns and invests in the region's electricity-generation capital, and a representative firm that produces the final good. Regions differ in wind and solar resources (Figure 2), electricity demand, initial generation capacity, and the interconnection within which they trade.

The economy transitions from observed 2010 capital stocks toward a stochastic steady state. The transition is driven by growing electricity demand, declining global technology costs, and federal renewable subsidies.

The federal subsidy is the model's only source of aggregate uncertainty. Subsidy availability follows a two-state Markov process $s_t \in \{0, 1\}$, where $s_t = 1$ is "subsidy on" and $s_t = 0$ is "subsidy off". Household-generators know the transition probabilities but not the dates of future policy changes.

Wind earns a production tax credit (PTC) of ϕ^{PTC} per MWh. Solar gets an investment

[7] The 13 regions are California (CAL), Carolinas (CAR), Central (CENT), Florida (FLA), Mid-Atlantic (MIDA), Midwest (MIDW), New England (NE), Northwest (NW), New York (NY), Southeast (SE), Southwest (SW), Tennessee (TEN), and Texas (TEX).

tax credit (ITC), ϕ^{ITC}, that reduces its up-front capital cost when the subsidy is active. Appendix A states the eligibility rules formally.

4.2 Households and generation investment

The representative household in region j consumes a final good C_{jt} and electricity E^{HH}_{jt}, supplies labor inelastically, saves in the national capital market, and owns the region's generation capital. It maximizes

$$\mathbb{E}_0 \sum_t \beta^t \log \mathcal{U}(C_{jt}, E^{HH}_{jt}), \qquad \mathcal{U}(C,E) = \left[C^{\frac{\sigma-1}{\sigma}} + \bar{\alpha}_h^{1/\sigma} E^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \tag{4.1}$$

where σ is the electricity-demand elasticity and $\bar{\alpha}_h$ is the electricity weight. I set σ using estimates from the empirical electricity-demand literature and, conditional on σ, calibrate $\bar{\alpha}_h$ so that electricity accounts for its observed 2% expenditure share (Section 5).

The household budget constraint is

$$C_{jt} + p^E_{jt} E^{HH}_{jt} + \sum_k \tilde{p}^k_t I^k_{jt} + \sum_k \Psi^k(I^k_{jt}, K^k_{jt}) + S_{jt} = (1-\varsigma_t) w_{jt} L_{jt} + r^K_t A_{jt} + \Pi^E_{jt} + \mathcal{R}_{jt} + N_j T^{nat}_t, \tag{4.2}$$

where S_{jt} is saving in the national capital market, Π^E_{jt} is electricity-sector profit, $\mathcal{R}_{jt}$ is the region's lump-sum share of interregional congestion revenues, and T^{nat}_t is a national lump-sum transfer. Financial wealth evolves according to

$$A_{j,t+1} = (1-\delta_P) A_{jt} + S_{jt}.$$

The labor-income tax ς_t finances federal subsidy payments and is non-distortionary because labor supply is inelastic.

The solar ITC lowers the effective price of solar investment,

$$\tilde{p}^S_t = (1 - \phi^{ITC} \chi^S_t) p^S_t,$$

while wind and gas pay their full capital prices, $\tilde{p}^W_t = p^W_t$ and $\tilde{p}^G_t = p^G_t$. The wind PTC instead enters electricity-sector profits as a payment on eligible wind generation. Investment is subject to convex adjustment costs $\Psi^k(I^k_{jt}, K^k_{jt})$, and wind faces a rising regional deployment-cost curve $c^W_j(\cdot)$ as a region builds out its most productive sites. Section 5.1

gives the functional forms and calibration.

Generation capital accumulates subject to irreversibility:

$$K^k_{j,t+1} = (1-\delta^k)K^k_{jt} + I^k_{jt}, \qquad I^k_{jt} \geq 0, \qquad k \in \{S, W, G\}. \tag{4.3}$$

To capture grandfathering of the wind PTC, I separately track subsidy-eligible wind capacity:

$$K^{W,PTC}_{j,t+1} = (1-\delta^W)(1-h^{exit})K^{W,PTC}_{jt} + I^W_{jt}\mathbf{1}[s_t = 1]. \tag{4.4}$$

Wind built while the subsidy is active enters both the physical and subsidy-eligible stocks; wind built during a lapse enters only the physical stock. The parameter h^{exit} governs the expiration of PTC eligibility.[8]

Let

$$\mathbf{K}_{jt} = (K^S_{jt}, K^W_{jt}, K^{W,PTC}_{jt}, K^G_{jt}).$$

The value of generation capital satisfies

$$\begin{aligned} V_t(\mathbf{K}_{jt}, s_t) = \max_{I^S, I^W, I^G \geq 0} \Big\{ & \Pi^E_{jt}(\mathbf{K}_{jt}, s_t) - \tilde{p}^S_t I^S_{jt} - \Xi^W_{jt}(I^W_{jt}) - \tilde{p}^G_t I^G_{jt} - \sum_k \Psi^k(I^k_{jt}, K^k_{jt}) \\ & + \mathbb{E}\left[\mathcal{M}_{t,t+1} V_{t+1}(\mathbf{K}_{j,t+1}, s_{t+1}) \mid s_t\right] \Big\}, \end{aligned} \tag{4.5}$$

subject to (4.3)–(4.4). Here $\Xi^W_{jt}(I^W) = \int_{K^W_{jt}}^{K^W_{jt}+I^W} c^W_j(k)\,dk$ is wind investment expenditure along the regional site-cost curve, and

$$\mathcal{M}_{t,t+1} = \beta \frac{\mathcal{U}_C(C_{j,t+1}, E^{HH}_{j,t+1})}{\mathcal{U}_C(C_{jt}, E^{HH}_{jt})}$$

is the household's stochastic discount factor.

The wind first-order condition illustrates how expectations about future subsidy availability affect investment. For $I^W_{jt} > 0$,

$$\underbrace{c^W_j(K^W_{jt} + I^W_{jt}) + \Psi^W_I}_{\text{marginal cost}} = \underbrace{\mathbb{E}[\mathcal{M} q^W_{t+1} \mid s_t]}_{\text{energy value}} + \mathbf{1}[s_t = 1] \underbrace{\mathbb{E}[\mathcal{M} q^{W,PTC}_{t+1} \mid s_t]}_{\text{subsidy value}}. \tag{4.6}$$

When the subsidy is active, a marginal unit of wind earns both its electricity-market value

[8] The model approximates the statutory ten-year PTC eligibility window with an annual exit rate for eligible capacity.

and the value of the grandfathered subsidy stream.

4.3 Production firms and the national capital market

A representative firm in region j produces the final good from capital, labor, and electricity. It combines a Cobb–Douglas capital–labor bundle with electricity in a constant-returns CES technology,

$$Y_{jt} = Z_{jt}\left[(1-\gamma_E)^{1/\sigma}\left(K_{P,jt}^{\hat{\alpha}_K}L_{jt}^{\hat{\alpha}_L}\right)^{\frac{\sigma-1}{\sigma}} + \gamma_E^{1/\sigma}\left(E_{jt}^{firm}\right)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}, \tag{4.7}$$

where

$$\hat{\alpha}_K = \frac{\alpha_K}{\alpha_K+\alpha_L}, \qquad \hat{\alpha}_L = \frac{\alpha_L}{\alpha_K+\alpha_L}.$$

Production capital is supplied through a national market. Households hold financial wealth A_{jt} and firms rent production capital $K_{P,jt}$ at the common rental rate r_t^K, which clears the national capital market:

$$\sum_j K_{P,jt} = \sum_j A_{jt}. \tag{4.8}$$

4.4 The electricity sector

The household-generator owns generation capacity in hydro, nuclear, wind, solar, gas, and coal. It chooses investment only in wind, solar, and gas. Hydro and nuclear are fixed stocks, while coal follows an exogenous retirement path and no new coal is built.

Dispatch. Generation *capacity* is distinct from electricity *generation*. Figure 6 illustrates the merit-order dispatch curve.

Within-period electricity demand follows a Pareto distribution with region-specific tail index θ_j and scale $x_{jt}^{\min}$,

$$\Pr(D_{jt} > d) = \left(\frac{x_{jt}^{\min}}{d}\right)^{\theta_j}, \qquad \mathbb{E}[D_{jt}] = \frac{\theta_j}{\theta_j - 1}x_{jt}^{\min}. \tag{4.9}$$

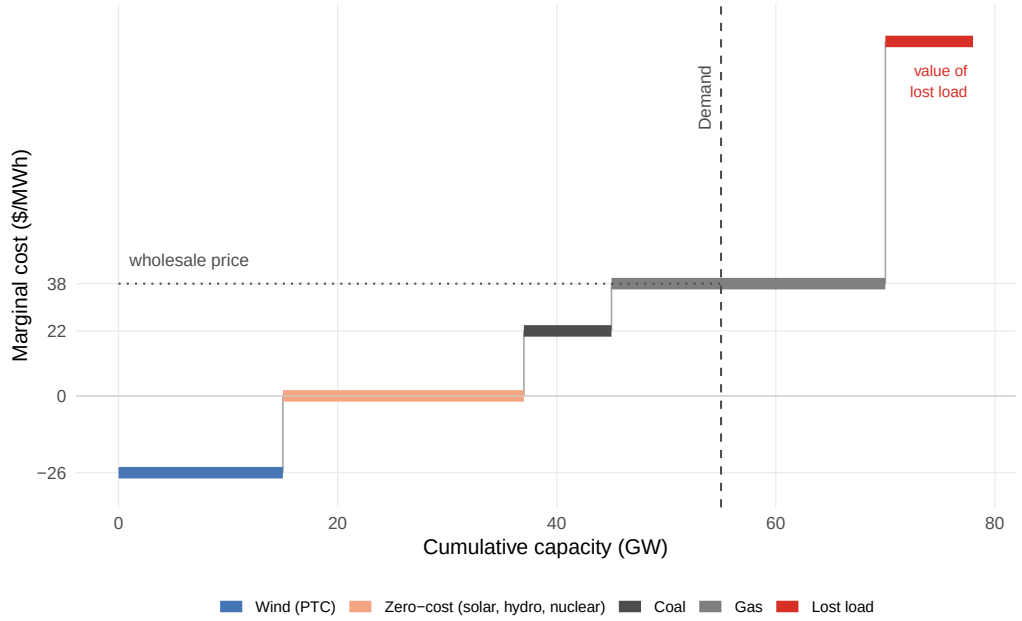


Figure 6: The merit-order dispatch curve.

Notes. Merit-order supply curve for a region in a single period. Illustrative capacities and costs.

A lower θ_j corresponds to a peakier load-duration curve. Annual household and firm electricity demand is price elastic,

$$E_{jt}^{firm} = \bar{\alpha}_E Y_{jt} (p_{jt}^E)^{-\sigma}, \qquad E_{jt}^{HH} = \bar{\alpha}_h C_{jt} (p_{jt}^E)^{-\sigma}, \tag{4.10}$$

and $x_{jt}^{\min}$ is chosen so that expected hourly demand aggregates to annual electricity demand.

The expected wholesale price is the expected marginal cost of serving load:

$$\begin{aligned} \bar{p}_{jt}^E = {} & mc_{jt}^{lo} \Pr(D_{jt} > Q_{jt}^0) \\ & + (mc_{jt}^{hi} - mc_{jt}^{lo}) \Pr(D_{jt} > Q_{jt}^0 + K_{jt}^{lo}) \\ & + (V_j - mc_{jt}^{hi}) \Pr(D_{jt} > \bar{Q}_{jt}), \end{aligned} \tag{4.11}$$

where Q_{jt}^0 is zero-marginal-cost capacity, mc_{jt}^{lo} and mc_{jt}^{hi} are the marginal costs of the cheaper and more expensive fossil technologies, and $\bar{Q}_{jt}$ is total capacity. Additional renewable capacity expands the zero-cost portion of the stack, makes fossil technologies marginal less often, and therefore lowers the electricity price earned by generation. Appendix A gives the corresponding closed-form expression under Pareto demand.

Capacity adequacy. I model a capacity-adequacy constraint which requires each region to maintain enough reliable generation capacity to meet forecast demand at a given loss-of-load-probability target. Because gas is the only dispatchable technology that can be built endogenously, it fills any capacity shortfall when existing baseline generation capacity is insufficient. The multiplier on the constraint is a capacity price p_{jt}^{cap}, which captures the value of additional reliable capacity. The requirement is based on each region's exogenous forecast demand. Appendix A.3 defines the constraint and capacity price formally.

Trade. Regions trade electricity within, but not across, the three interconnections. Let $X_{jt} = G_{jt} - \bar{E}_{jt}$ denote region j's net exports. The local electricity price is the interconnection hub price net of a congestion basis,

$$p_{jt}^{E} = \bar{p}_{mt}^{E} - \gamma \frac{X_{jt}}{\overline{TL}_j}, \tag{4.12}$$

where $\overline{TL}_j$ measures transmission capacity. An exporter's local price falls relative to the hub price as congestion increases, while an importer pays a higher local price. Each interconnection's hub price clears aggregate net trade among its member regions. Transmission costs are real resource costs; congestion-basis payments are transfers and are rebated lump-sum through $\mathcal{R}_{jt}$. Appendix A gives the complete electricity-sector profit and trade accounting.

National electricity-sector CO_2 emissions are

$$E_t = \sum_j \left(\xi^G gen_{jt}^G + \xi^C gen_{jt}^C \right). \tag{4.13}$$

4.5 Equilibrium

Given the exogenous paths and policy transition matrix Π, a recursive competitive equilibrium is a sequence of household allocations, generation and production capital, investment policies, electricity generation and trade, and prices such that: *(i)* households optimize consumption, saving, and generation investment; *(ii)* final-good firms maximize profits; *(iii)* generation is dispatched in merit order and electricity markets clear within each interconnection; *(iv)* the national capital market and regional labor markets clear; *(v)* the federal budget and aggregate resource constraint hold; and *(vi)* expectations are consistent with the policy transition matrix Π. The economy converges to a terminal stochastic steady state. Appendix A states the equilibrium conditions formally, and Appendix B describes the

computational solution.

4.6 Welfare

I measure welfare as the present value of real consumption net of the environmental damages from power-sector emissions. The environmental damages have two components: global climate damages from CO_2 and local air-pollution mortality damages from fossil generation,

$$\begin{aligned} W^{C} &= \sum_{t} \beta^{t} \sum_{j} C_{jt}, \qquad W^{\mathrm{CO_2}} = \sum_{t} \left(\frac{1}{1+r_c}\right)^{t-2010} \mathrm{SCC}_t \, E_t, \\ W^{\mathrm{AP}} &= \sum_{t} \left(\frac{1}{1+r}\right)^{t-2010} \sum_{j} \left(d_{jC}\, gen^{C}_{jt} + d_{jG}\, gen^{G}_{jt}\right), \end{aligned} \tag{4.14}$$

and total welfare is $W^{\$} = W^{C} - W^{\mathrm{CO_2}} - W^{\mathrm{AP}}$. Consumption is discounted at the household rate β. The climate component values each year's CO_2 emissions E_t at the EPA social cost of carbon, discounted at 2% (U.S. Environmental Protection Agency, 2023). The local air-pollution component values regional coal and gas generation gen^{C}_{jt} and gen^{G}_{jt} at regional fuel-specific marginal damage coefficients d_{jC} and d_{jG}, and discounts the resulting contemporaneous mortality damages at the private rate r. The marginal damage coefficients come from the AP4 integrated assessment model and value $PM_{2.5}$-related mortality; Appendix F.3 gives their construction.

Subsidies, the taxes that finance them, household electricity payments, and congestion rebates are transfers and therefore cancel from aggregate welfare. Generation investment, adjustment costs, fossil fuel use, transmission, and production-capital accumulation are real resource costs. Appendix A.5 gives the equivalent aggregate resource constraint.

5 Calibration

I calibrate the model in three steps. First, I set technology, demand, emissions, and regional electricity-market parameters using external data. Second, I calibrate the investment and site-cost parameters to reproduce national U.S. wind and solar capacity levels over 2010–2024. Third, I calibrate firms' expectations about future subsidy availability using the historical policy record and transitions in federal political control. Tables 2 and 3 summarize

the data sources and parameter values.

5.1 External parameters and exogenous paths

The exogenous paths come from standard energy datasets. Wind, solar, and gas capital costs follow the NREL Annual Technology Baseline (Moderate) projection, joined to historical overnight costs from LBNL and EIA. Electricity demand follows the EIA Annual Energy Outlook 2025 reference case. Fuel costs and the exogenous coal-retirement path come from EIA. Appendix Figure 12 plots these paths.

Regional renewable productivity is measured using realized capacity factors, calculated as net generation from EIA-923 divided by nameplate capacity from EIA-860 and averaged over 2019–2024. I use NREL developable-resource data to calibrate the regional scale of wind and solar site availability. I use EIA-930 to calculate the shape of within-year electricity demand and calibrate the parameters for interregional trade. Appendix E describes the construction of each input and Table 10 reports regional summary statistics.

The remaining externally calibrated parameters come from engineering estimates and the empirical literature. Technology-specific depreciation rates and renewable capacity credits come from NREL, emissions factors from EPA eGRID, and production cost shares from BEA. The wind PTC and solar ITC are set to their legislation values. Panel A of Table 3 reports the full set of externally calibrated parameters.

5.2 Internal calibration

Four parameters govern how quickly renewable investment responds to changes in profitability. Investment adjustment costs are quadratic in the investment rate,

$$\Psi^k(I, K) = \frac{\psi^k}{2}\left(\frac{I}{K}\right)^2 K,$$

and capture the increasing cost of expanding generation capacity rapidly. Wind deployment costs also rise as a region exhausts its most productive sites:

$$c_j^W(K_{jt}^W) = \tilde{p}_t^W\left[1 + \left(\frac{K_{jt}^W}{\bar{K}_j^W}\right)^\eta\right], \tag{5.1}$$

Table 2: Data sources.

Dataset	Source	Years	Use in the model
Existing capacity	EIA-860	2010–2024	Initial stocks $K^k_{j,0}$, coal-retirement path
Generation & fuel	EIA-923	2010–2024	Capacity factors CF^k_j; fuel marginal costs
Hourly demand & interchange	EIA-930	2021–2023	Pareto demand shape θ_j; trade capacity $\overline{TL}_j$
Electricity demand path	EIA AEO 2025 Ref.	2024–2050	Demand growth (electrification growth)
Capital-cost projection	NREL ATB 2024 (Moderate)	2025–2050	Renewable and gas cost decline
Developable resource	NREL land-based resource	2023	Site-cost scales $\bar{K}^k_j$
Wholesale prices	ICE, NYISO	2021–2025	Interconnection trade slope s_j
Emissions factors	EPA eGRID	2022	ξ_G, ξ_C
Household counts	U.S. Census	2020	Regional L_j; state-to-region aggregation
Policy history	PTC/ITC statutes, IRS notices	1992–2025	Eligibility; forward Markov chain
Region geography	EIA / Census TIGER	2019	13-region map
Local air-pollution damages	AP4 / EIA-923	2017	$PM_{2.5}$-related mortality damage coefficients

Notes. The AEO 2025 demand path and the NREL ATB 2024 cost projection are published through 2050 and held (or extrapolated along their terminal trend) over the remaining model horizon.

where $\bar{K}^W_j$ is proportional to region j's developable wind resource and I set $\eta = 2$.

I calibrate the wind and solar adjustment-cost parameters and the wind and solar site-cost scales jointly to reproduce national wind and solar capacity levels over 2010–2024. Adjustment costs determine how quickly investment can respond when profitability changes. The site-cost scales determine how rapidly increasingly expensive locations are reached as cumulative deployment rises.

5.3 Policy expectations

The availability of the subsidies follows a two-state Markov process,

$$s_t \in \{0, 1\},$$

with π_{11} denoting the probability that an active subsidy remains active and π_{00} the probability that a lapse continues for another year.

Table 3: Model parameters.

Parameter	Symbol	Value
Panel A. Externally calibrated (data, engineering, tax code)		
Discount factor	β	0.9615 ($\rho \approx 4\%$)
Wind, solar depreciation	δ^W, δ^S	0.04, 0.033 (NREL ATB)
Gas, production-capital depreciation	δ^G, δ_P	0.03, 0.05
PTC value	ϕ^{PTC}	\$26/MWh (§45)
ITC rate	ϕ^{ITC}	30% (§48)
Solar capital cost path	p_t^S	\$4.27M→\$0.79M/MW
Wind capital cost path	p_t^W	\$2.46M→\$1.10M/MW
Gas capital cost path	p_t^G	\$1.33M→\$0.74M/MW
Elasticity	σ	0.3
Emissions factors	ξ^G, ξ^C	0.40, 0.95 tCO_2/MWh (eGRID)
Social cost of carbon	SCC	\$213/tCO_2 (EPA 2023)
PTC vintage duration	h^{exit}	0.1 (§45)
Site-cost curvature	η	2
Capacity factors	CF_j^k	wind 0.04–0.38, solar 0.19–0.32
Trade slope (congestion)	$s_j = \gamma/\overline{TL}_j$	0.24–8.6, $\gamma \approx 2$ (ICE/NYISO, EIA-930)
Reliability target	ε	0.002 (NERC)
Capacity credit	ζ^W, ζ^S	0.15, 0.10 (NREL)
Cost shares (capital/labor/electricity)	$\alpha_K, \alpha_L, \alpha_E$	0.33, 0.65, 0.02 (BEA)
Electricity demand weights	$\bar{\alpha}_h, \bar{\alpha}_E$	$0.0014 = 0.02 \times 45^{-0.7}$ (2% share)
Panel B. Internally calibrated to the 2010–2024 capacity growth		
Wind, solar adjustment cost	$\bar{\psi}_W, \bar{\psi}_S$	2×10^5 each
Gas adjustment cost	$\bar{\psi}_G$	10^7
Wind site-cost scale	κ_W	0.08 (2024 wind ≈143 GW)
Solar site-cost scale	κ_S	0.22 (2024 solar ≈136 GW)

Notes. Panel A is set from external data, engineering studies, and the tax code. Panel B's four deployment-cost parameters are calibrated to the observed 2010–2024 renewable capacity growth, matching the 2024 capacity levels. The policy Markov chain is calibrated separately (Section 5.3).

I use the history of federal political control to discipline π_{00}. I classify each Congress since 1993 as a Democratic trifecta, a Republican trifecta, or divided government and estimate transitions among the three states (Table 9). I map a Democratic trifecta into the active subsidy state and a Republican trifecta into the lapsed state.

I convert the estimated political process to annual transition probabilities by preserving its expected spell lengths. This gives $\pi_{11} = 0.90$ (an expected ten years active) and $\pi_{00} = 0.94$ (an expected seventeen-year lapse). The robustness analysis also considers faster renewal and permanent repeal. Section 7 shows how the results change across this range.

5.4 Validation: untargeted moments

The calibration uses national wind and solar capacity levels from 2010–2024, but does not target emissions, generation shares, electricity prices, the location of renewable capacity, or electricity trade.

At the national level, the model produces 1.52 Gt of power-sector CO_2 emissions in 2024, compared with 1.5 Gt in the data. Wind supplies 10% of generation in both the model and the data, hydro and nuclear supply 25% in the model and 24% in the data. Fossil fuel together supply 57% and 58%, respectively. The main difference is solar: its share is 8% in the model and 5% in the data. The model's average wholesale electricity price also falls within the observed 2021–2024 range.

The regional allocation of renewable capacity is not a calibration target. However, the Spearman rank correlation between modeled and observed wind capacity across the 13 regions in 2024 is $\rho = 0.96$. The match is weaker for solar, with $\rho = 0.57$. The model gets the direction of net electricity trade right in 8 of the 12 regions that trade across regional boundaries (excluding Texas as it does not trade).

Figures 14 and 15 provide the corresponding dynamic and regional comparisons.

Table 4: Untargeted validation: model versus data.

Moment	Model	Data
Panel A. Aggregate moments		
Power-sector CO_2, 2010 (Gt/yr)	2.15	2.2
Power-sector CO_2, 2024 (Gt/yr)	1.52	1.5
Wholesale price, 2021–2024 ($/MWh)	41	35–45
Wind generation share, 2024 (%)	10	10
Solar generation share, 2024 (%)	8	5
Baseload generation share, 2024 (%)	25	24
Fossil generation share, 2024 (%)	57	58
Panel B. Spatial agreement		
Wind capacity across regions, Spearman ρ	0.96	
Solar capacity across regions, Spearman ρ	0.57	
Net-trade sign	8/12 regions	

Notes. Regional capacity statistics report the Spearman rank correlation between modeled and observed 2024 capacity across the 13 regions. The trade statistic reports the number of regions for which the model reproduces the observed sign of net trade; Texas is excluded because ERCOT has no interconnection trade in the model.

6 Counterfactuals

All counterfactuals share the realized 2010–2027 policy history and therefore enter 2028 with identical state variables.[9] The policy experiments branch in 2028, when the OBBBA repeal takes effect.

I consider three main policy regimes: a permanent subsidy, an uncertain subsidy, and no subsidy.

6.1 Policy regimes

Permanent subsidy. The subsidy remains available in every period, $s_t = 1$. Wind receives the production tax credit of $26/MWh and solar receives the 30% investment tax credit. Firms know that the subsidy will remain available when making investment decisions. The present value of government spending under this policy is $231 billion.

[9]The model transition begins in 2010 and is calibrated and validated against the observed electricity transition. The 2010 start date does not affect differences across the post-2028 policy experiments.

Uncertain subsidy. The subsidy availability follows the two-state Markov process calibrated in Section 5.3. The transition probabilities are known but the dates of future lapses and renewals are not. I keep the solar ITC at 30% and set the active-state wind PTC to \$36.4/MWh, so that expected present-value subsidy spending under the uncertain policy also equals \$231 billion.

No subsidy. The subsidy is permanently unavailable, $s_t = 0$, and firms know that it will not return.

6.2 Counterfactual comparisons

My main welfare comparison is

$$W^{\text{perm}} - W^{\text{unc}},$$

which I call the *value of continuous subsidy*. The permanent and uncertain policies have the same expected present-value subsidy spending.

I also report

$$W^{\text{unc}} - W^{\text{none}},$$

the value of the uncertain subsidy relative to no federal subsidy. The two differences add up to the value of permanent support relative to no subsidy:

$$W^{\text{perm}} - W^{\text{none}} = (W^{\text{perm}} - W^{\text{unc}}) + (W^{\text{unc}} - W^{\text{none}}).$$

6.3 Perfect-foresight counterfactuals

To separate uncertainty about future policy dates from the effects of changes in subsidy availability themselves, I construct a full-foresight counterfactual. I draw subsidy paths from the same Markov process that governs the stochastic policy in the baseline. For each realized path, I reveal the complete future subsidy schedule to firms in 2028 and solve the model under perfect foresight. Averaging outcomes across these paths gives expected welfare under the same distribution of future subsidy realizations as the baseline, but without uncertainty about when policy changes occur.

This experiment decomposes the value of continuous subsidy into two components. Let W^{perm} denote welfare under continuous support, W^{unc} welfare under the stochastic policy, and $\mathbb{E}_\omega[W^{\text{PF}}(\omega)]$ expected welfare when each realized policy path ω is known in advance. Then

$$W^{\text{perm}} - W^{\text{unc}} = \left(W^{\text{perm}} - \mathbb{E}_\omega[W^{\text{PF}}(\omega)]\right) + \left(\mathbb{E}_\omega[W^{\text{PF}}(\omega)] - W^{\text{unc}}\right).$$

The first term captures the value of continuous subsidy when future policy dates are known, while the second captures the value of resolving uncertainty about the future policy path.

I also consider two deterministic paths to illustrate how investment responds to the timing and duration of known subsidy lapses. In the *short-lapse* scenario, the subsidy is off from 2028 through 2033, on from 2034 through 2043, and off again beginning in 2044. In the *long-lapse* scenario, the subsidy is off from 2028 through 2044, on from 2045 through 2054, and off again beginning in 2055. Household-generators know the relevant schedule in 2028 and solve their investment problem using that known sequence rather than expectations over future subsidy states.

I adjust the active-state wind PTC in each deterministic foresight path so that present-value subsidy spending is equal to spending under continuous support. The continuous-support policy spends $230.9 billion in present value, compared with $232.0 billion under the short-lapse path and $232.3 billion under the long-lapse path. The deterministic foresight experiments therefore compare continuous and known on-and-off support at essentially the same fiscal cost while removing uncertainty about future policy dates.

7 Results

7.1 The value of budget-neutral continuous subsidy

At the same expected budget, continuous subsidy produces more renewable capacity and less gas than the stochastic on-and-off subsidy. By 2070, continuous support has 137 GW more renewable capacity and 37 GW less gas (Figure 7). The lower renewable capacity under the stochastic policy translates into higher fossil generation and emissions (Figure 8). Cumulatively, the stochastic on-and-off policy produces 2.29 Gt more CO_2 through 2070, or 3.2% relative to continuous support, equivalent to roughly a year and a half of U.S. power-sector emissions in 2024.

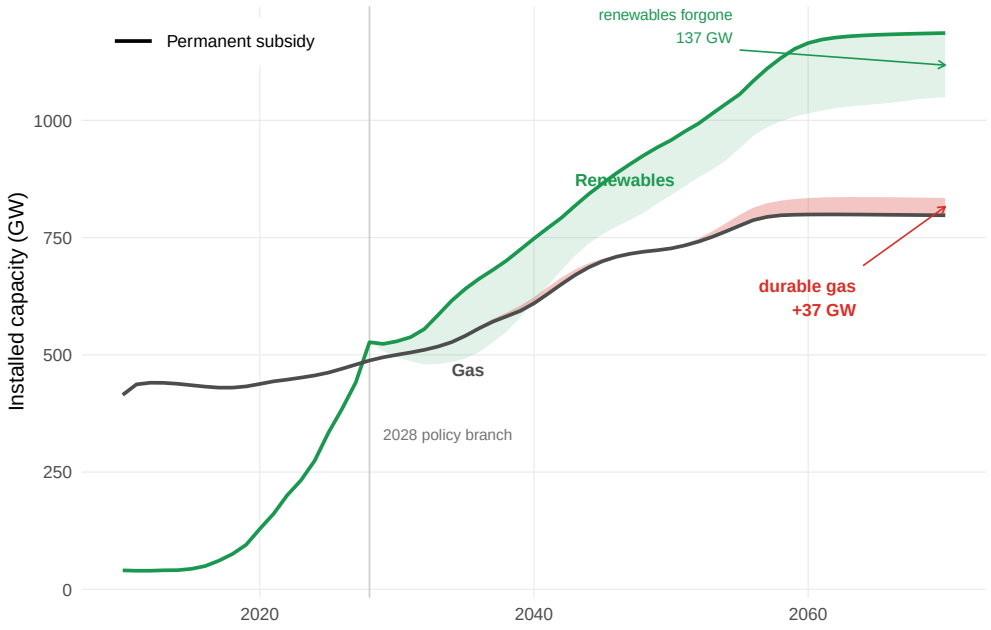


Figure 7: **Renewable and gas capacity under continuous and budget-neutral stochastic support.**

Notes. Installed renewable (wind plus solar) and gas capacity under continuous support (solid) and the budget-neutral stochastic subsidy (dashed).

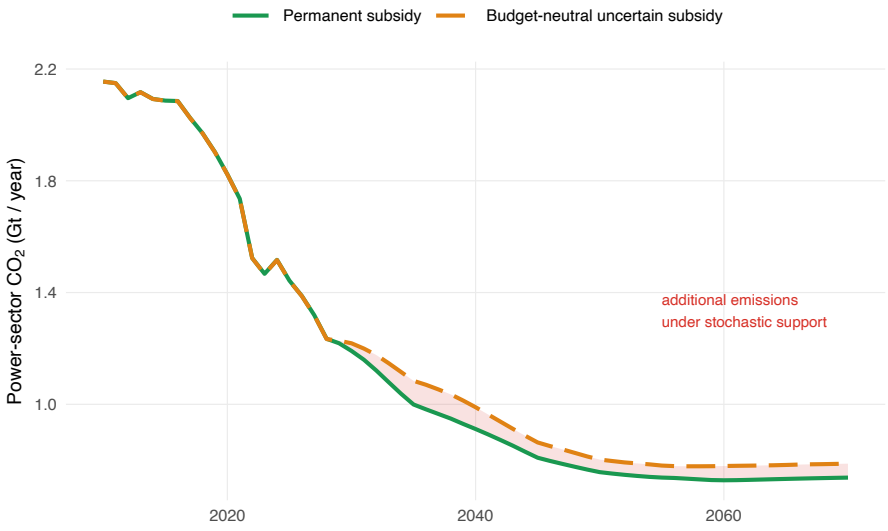


Figure 8: Annual power-sector CO_2 by policy regime.

Notes. Annual U.S. power-sector CO_2 emissions under continuous support and the budget-neutral stochastic subsidy.

Most of the emissions difference does not come from additional gas capacity. Existing fossil plants generate more when renewable investment is delayed. Additional generation from existing coal and gas accounts for 1.83 Gt, or 80% of the cumulative emissions gap. Of

this component, 52% comes from existing gas and 48% from coal. The remaining 0.46 Gt (20%) comes from additional gas capacity built under the stochastic policy.

Table 5: The value of budget-neutral continuous subsidy, decomposed.

($B, present value)	Permanent − uncertain	Uncertain − no subsidy
Environmental gains	**406**	**227**
Climate damages	360	203
existing coal	177	95
existing gas	116	63
new gas	66	45
Local air-pollution mortality	46	24
coal	39	20
gas	7	4
Consumption (net change)	**−113**	**−134**
Renewable & baseload income	144	66
Wages	16	12
Gas & coal profit	−23	−9
Generation investment	−251	−203
Value	**292**	**93**
Value excluding local air pollution	246	69

Notes. Present value in billions of real 2022 dollars, horizon through 2070. Consumption is the net change in the present value of consumption; income, wages, and profit raise it, while generation investment enters negatively. Value = environmental gains + net change in consumption. Column 1 is the value of continuous support (versus a budget-neutral uncertain subsidy); column 2 is the value of the uncertain subsidy versus no subsidy.

Table 5 decomposes the welfare change. Replacing the stochastic subsidy with budget-neutral continuous support generates $292 billion in net benefits. Continuous support avoids $406 billion in environmental damages: $360 billion in climate damages from CO_2 and $46 billion in mortality damages from local air pollution. Most of the climate benefit comes from lower generation by existing coal and gas, with the remainder from gas capacity that is never built.

These benefits are partly offset by a consumption cost. Continuous support adds more renewable investment and lowers consumption-equivalent welfare by 0.12%, a present-value consumption cost of $113 billion (Table 5). Excluding local air-pollution damages, the value of continuous support is $246 billion; valuing only avoided CO_2, continuous support breaks even at a social cost of carbon of $\$67/tCO_2$.

Including avoided local air-pollution mortality raises the value of the stochastic subsidy relative to no subsidy from $69 billion to $93 billion. The value of continuous subsidy remains

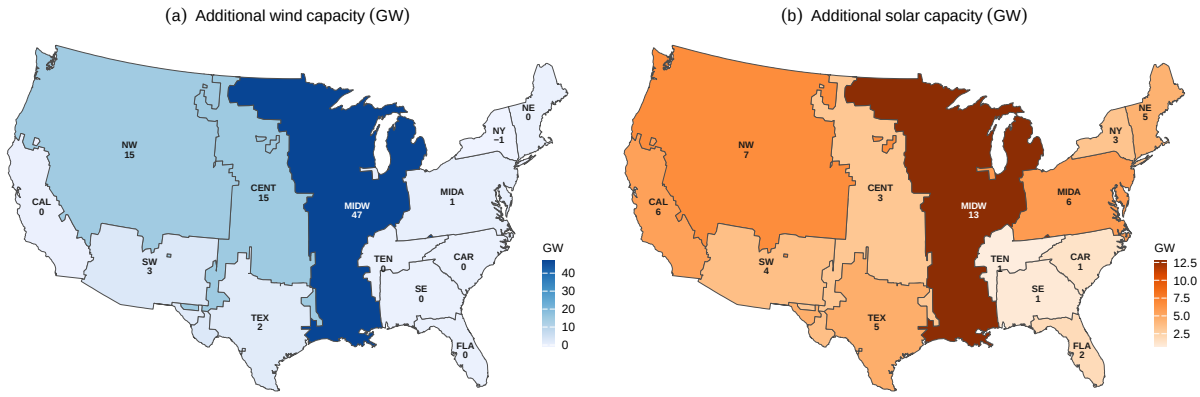


Figure 9: Regional capacity response to continuous subsidy.

Notes. Additional installed capacity under continuous support relative to the budget-neutral stochastic subsidy in 2070, by EIA region. Panel (a) reports wind and panel (b) solar.

positive under faster subsidy renewal, permanent repeal, alternative reporting horizons, and discounting climate damages at the market rate, ranging from $56 billion to $385 billion ($48–$315 billion excluding local air pollution); Appendix G reports these exercises.

7.2 Regional heterogeneity

The aggregate investment response is unevenly distributed across regions (Figure 9). The wind response is concentrated in the interior: the Midwest, Central, and Northwest together build 76 GW more wind under continuous support. In contrast, regions with poor wind resources, such as Tennessee and Florida, build little additional wind. The solar response is more geographically dispersed. Continuous support increases solar capacity across nearly every region, with the largest increase in the Midwest and substantial responses in the Northwest, California, the Mid-Atlantic, New England, and Texas. The Midwest has the largest combined wind and solar capacity response.

The effect of continuous support is largest neither in regions with the best renewable resources nor in those with the worst. Where renewable projects are profitable without federal support, investment can continue during a lapse. At the other extreme, where renewable resources are poor, even the subsidy may not be sufficient to make additional projects profitable. Continuous support matters most for the marginal regions, where the subsidy determines whether additional renewable capacity is built or not.

The emissions response also depends on the generation capacity that was already in

place when the policies diverge in 2028. When renewable investment is delayed, regions with substantial existing fossil capacity can meet more electricity demand by running those plants more intensively. The Midwest has the largest response due to large initial carbon-intensive fossil capacity (especially coal) which generates additional emissions when the renewable investment is delayed.

The environmental consequences depend not only on how much fossil generation responds, but also on which fossil technologies respond. This distinction is particularly important for local air pollution. Using plant-specific marginal damages from AP4, the generation-weighted $PM_{2.5}$-related mortality damage of coal generation is approximately \$105/MWh nationally, compared with \$11/MWh for gas (in 2020 dollars). Regions in which continuous support reduces coal generation therefore account for a disproportionate share of the local air-pollution benefit.

Electricity trade creates an additional source of spatial spillovers: renewable investment in one region can reduce fossil generation elsewhere. Regions trade within the Eastern and Western interconnections, while Texas forms a separate interconnection. California illustrates this mechanism: it is a net importer, and its own change in renewable-capacity is relatively small. Under continuous support, however, the Northwest and Southwest (which both have a high solar suitability) build more renewable capacity and supply more clean electricity within the Western Interconnection. California imports part of this additional generation and runs its own gas capacity less.

Table 6 reports the corresponding regional decomposition. I assign the climate component according to the region in which power-sector CO_2 emissions change. Because CO_2 damages are global, this allocation is an accounting decomposition of the national climate benefit rather than a measure of local environmental benefits or regional welfare incidence. Avoided local air-pollution mortality damages are attributed to the source region of the emitting plants; Appendix F.3 describes their construction.

The regional decomposition confirms the importance of the existing fossil fleet. Of the Midwest's \$129 billion in climate gains under continuous support, \$98 billion comes from reduced generation by existing coal and another \$17 billion from existing gas; only \$13 billion comes from avoided new gas. The concentration of the response in coal also makes the Midwest particularly important for local air pollution. Plants located in the Midwest account for approximately \$25 billion, or 54%, of national avoided $PM_{2.5}$-related mortality damages (Appendix F.3). These damages are attributed to the location of the emitting plants and do

Table 6: The value of continuous support by region, decomposed (continuous support versus budget-neutral uncertain subsidy).

	Climate gains			Local poll.†	Consumption					Total
Region	Coal	Gas	Total	Avoided	Invest.	Income	Wages	Profit	Total	value
Midwest	98	30	129	25	−59	66	−3	−4	0	154
Texas	38	8	45	12	−76	72	−8	1	−12	46
California	0	54	54	1	−24	10	4	−7	−17	38
Mid-Atlantic	0	29	29	2	−16	7	0	0	−9	23
Tennessee	0	4	4	0	7	1	0	0	8	12
New England	0	17	17	0	−12	6	0	−1	−7	11
Florida	1	14	14	1	−8	3	0	0	−5	10
New York	0	11	11	0	−8	4	1	−2	−5	6
Carolinas	6	0	7	1	−3	0	1	0	−2	5
Southeast	1	3	4	0	−2	1	0	0	−2	2
Northwest	20	3	24	1	−19	−13	13	−7	−26	0
Southwest	8	3	11	1	−13	−3	4	−2	−14	−2
Central	5	5	10	1	−17	−9	4	0	−23	−12
National	**177**	**183**	**360**	**46**	**−251**	**144**	**16**	**−23**	**−113**	**292**

Notes. Present value in billions of real 2022 dollars, by region, sorted by total value. Climate gains value the region's avoided CO_2 at the EPA year-specific social cost of carbon discounted at 2%, split into existing coal and total gas. Consumption is the net change in the present value of the region's consumption; income, wages, and profit raise it, while generation investment enters negatively. Total value = climate gains + local air pollution + consumption; every column sums to the National row, which matches Table 5. †Avoided local air-pollution ($PM_{2.5}$) mortality damages attributed to emissions from fossil plants *located in* the region (AP4, discounted at 4%). Because pollution disperses across regions, much of the avoided mortality occurs downwind, so these are not health benefits received by residents of the source region.

not measure health benefits received specifically by Midwest residents.

7.3 Uncertainty versus known policy switches

The full-foresight counterfactual confirms that knowing future policy dates does not eliminate the value of continuous subsidy. Across realized policy paths, firms bunch investment toward periods in which the subsidy is available. This reduces the resource cost of adjusting investment across policy states, but long subsidy lapses still leave renewable capacity lower and fossil generation higher than under continuous support. The environmental value of continuous subsidy therefore remains substantial even when future policy dates are known.

The deterministic paths illustrate how firms adjust to known policy changes. Figure 10 shows the short-lapse path. Renewable investment falls during the 2028–2033 lapse, jumps when eligibility returns in 2034, and rises again before the known 2044 lapse. Construction becomes concentrated inside the known eligibility window as firms anticipate both its beginning and its end which is costly because of adjustment costs, while electricity demand continues to grow during the lapse.

Table 7 reports the welfare effects of the two illustrative paths. The short-lapse policy spends $232.0 billion in present value, compared with $230.9 billion under continuous support, yet continuous support is worth $273 billion on this path. The value rises to $445 billion under the long-lapse path. The local air-pollution component also rises from $28 billion to $80 billion because the longer lapse induces substantially more coal generation. Excluding local air pollution, the corresponding values are $245 billion and $365 billion.

Table 7: Continuous subsidy versus deterministic perfect-foresight policies.

Policy	First lapse (years)	PV subsidy spending ($B)	Value of cont. support ($B)	Excl. local air poll. ($B)
Continuous support	—	230.9	—	—
Perfect foresight: short lapse	6	232.0	273	245
Perfect foresight: long lapse	17	232.3	445	365

Notes. The value of continuous subsidy includes avoided climate damages and avoided local air-pollution mortality damages, in real 2022 dollars; the final column excludes local air pollution. The local-pollution component is $28 billion for the short-lapse path and $80 billion for the long-lapse path. Valuing the change in gas generation at the recent combined-cycle rate ($4.16/MWh) gives $271 billion and $440 billion. In the short-lapse path the subsidy is off in 2028–2033, on in 2034–2043, and off beginning in 2044; in the long-lapse path it is off in 2028–2044, on in 2045–2054, and off beginning in 2055.

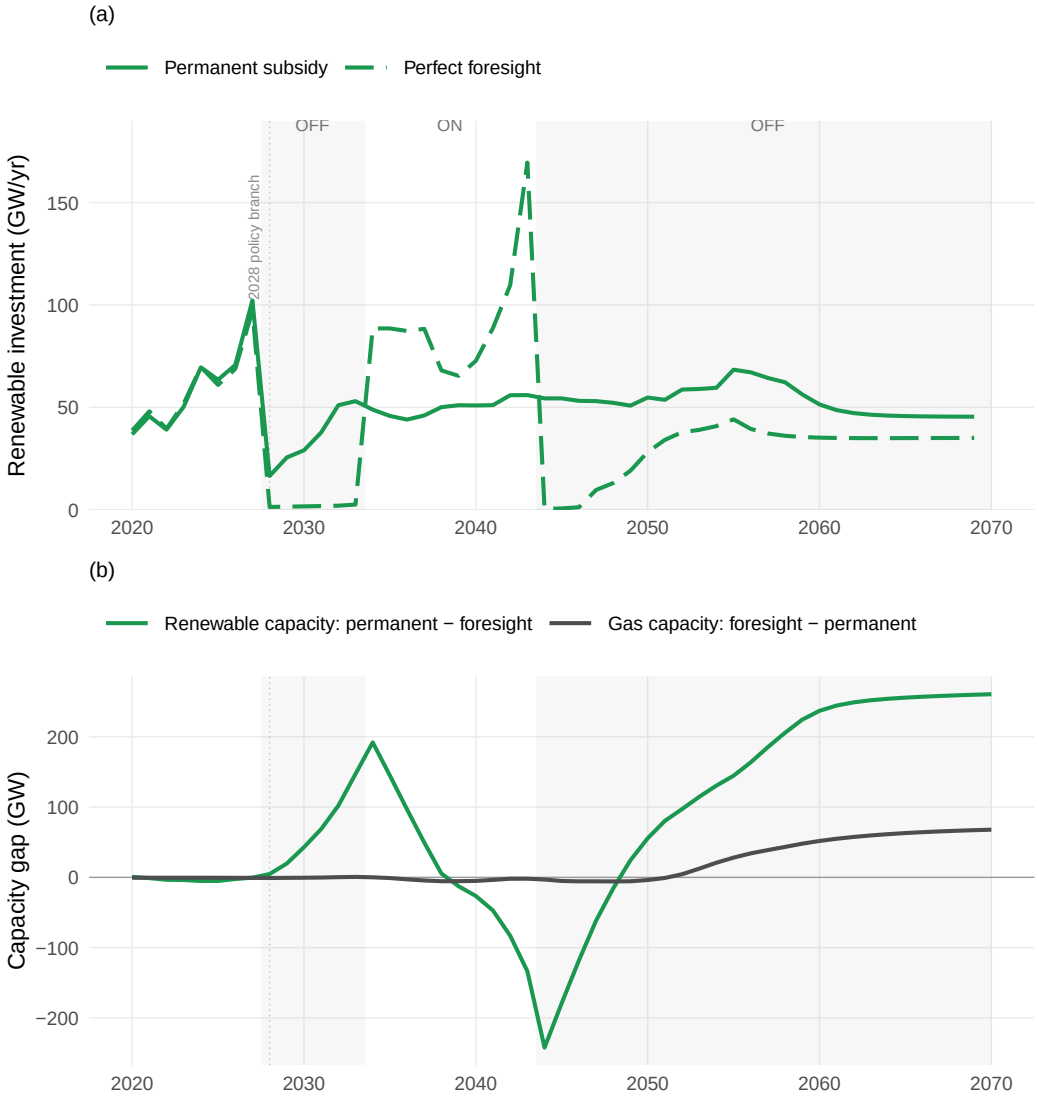


Figure 10: Investment and capacity under perfect foresight: short lapse.

Notes. Firms learn the complete future subsidy schedule in 2028. The subsidy is off in 2028–2033, on in 2034–2043, and off beginning in 2044. Panel (a) reports annual renewable investment under continuous support and the deterministic perfect-foresight policy. Panel (b) reports the renewable-capacity gap as continuous support minus perfect foresight and the gas-capacity gap as perfect foresight minus continuous support. Shaded areas indicate periods in which the subsidy is unavailable.

Lapse duration matters because foresight cannot make investment costlessly transferable across time. A short anticipated lapse gives firms more scope to bunch investment toward eligible years and to rely on subsidy-eligible wind grandfathered before the lapse. A longer interruption leaves less scope for this intertemporal substitution. Renewable capacity consequently remains lower for longer, and fossil generation fills more of the gap.

7.4 Discussion

The model abstracts from several mechanisms and margins that could affect the magnitude, timing, and spatial distribution of the response to changes in federal renewable subsidies. I discuss the main ones here.

Coal retirements. Coal retirements are exogenous in the model. In practice, utilities can postpone planned retirements when additional generation is needed. During a subsidy lapse, delayed renewable investment could therefore lead utilities to keep existing coal plants operating for longer. Allowing endogenous retirement could increase reliance on existing coal generation while renewable investment is low and reduce the need for additional gas capacity. Because coal is more emissions-intensive than gas, the effect on cumulative emissions is ambiguous even if less new gas capacity is built.

Gas construction lags. New gas capacity can be built within one period in the model, while in practice gas projects face permitting and construction lags. Longer construction times would limit the ability to build gas capacity in response to delayed renewable investment and shift more of the short-run response toward greater utilization of the existing fossil fleet. They could also reduce the persistence of the capital-stock difference, because less additional gas capacity would be built during a subsidy lapse. In the baseline counterfactual, 80% of the cumulative emissions difference already comes from greater utilization of existing coal and gas rather than additional gas capacity.

Learning-by-doing. The model does not include learning-by-doing. If additional U.S. deployment lowers subsequent renewable costs, maintaining the subsidy could add more current investment and lower future investment costs, amplifying the difference in deployment across policy paths. The magnitude of this channel depends on how strongly technology costs respond to U.S. deployment rather than global manufacturing and deployment.

Storage. The model has no grid storage. Storage complements renewables by shifting electricity from low-price to high-price hours. Including it could affect the continuous-support

result in two directions. In some regions, storage could make renewable projects profitable without federal support, allowing investment to continue during a lapse and narrowing the gap between the permanent and stochastic paths. In regions where investment would still stop during a lapse, however, the forgone renewable capacity would be more valuable, widening the gap. The net effect is therefore ambiguous.

Intraday renewable generation. Wind and solar generation are represented using annual capacity factors. The model therefore does not capture the covariance between renewable output and electricity demand or differences in the timing of wind and solar generation. An hourly representation could raise the value of solar where output coincides with peak demand, but lower it as midday generation becomes abundant and wholesale prices fall. These effects would also interact with storage. Incorporating hourly renewable profiles would refine the technology mix and dispatch response, although the policy mechanism studied here would not be affected.

Nuclear generation. Nuclear capacity is fixed in the model. Allowing new nuclear investment would provide another source of firm low-carbon capacity and could reduce the gas investment induced by slower renewable deployment.

Electricity-market institutions. Each region is represented by a common generation-investment problem, abstracting from differences among investor-owned utilities, publicly owned utilities, and merchant generators, as well as between vertically integrated and restructured electricity markets. These institutions can affect financing, cost recovery, exposure to wholesale electricity prices, and the response of investment to federal tax credits. The model captures geographic heterogeneity in renewable resources, demand, the existing generation fleet, and electricity trade, but not heterogeneity in ownership and regulation within regions. Incorporating these institutional differences could therefore change the magnitude and geographic distribution of the investment response.

State clean-electricity policies. The model abstracts from state renewable portfolio standards and clean-electricity standards. State policies can partly substitute for federal incentives when federal support lapses, reducing the response to federal subsidy availability in states with binding clean-energy requirements. Because these policies vary substantially across states, incorporating them could change both the aggregate response and its spatial distribution.

8 Conclusion

Federal renewable subsidies encourage investment in wind and solar, but their future availability has historically been uncertain. This paper quantifies the aggregate consequences of uncertainty about federal renewable subsidies in the U.S. electricity sector. At the same expected fiscal cost, continuous support produces 13% more renewable capacity than a stochastic subsidy that switches on and off and reduces investment in gas. The resulting reduction in fossil generation avoids $360 billion in climate damages and $46 billion in local air-pollution mortality damages. Accounting for the additional resources devoted to renewable investment, continuous support generates $292 billion in net benefits.

The aggregate cost arises because policy changes affect both when new capacity is built and how existing power plants are used. During subsidy lapses, renewable developers delay investment while electricity demand continues to grow. Existing coal and gas plants generate more in the meantime, accounting for 80% of the cumulative emissions difference, while the remaining 20% comes from additional gas capacity. Renewable investment can resume when support returns, but gas capacity built during a lapse is irreversible and remains in the generation stock until it depreciates. Changes in policy availability can therefore affect emissions both while support is unavailable and after it returns.

The effects also differ substantially across regions. Continuous support matters most where federal subsidies change the renewable investment decision and existing fossil capacity can fill the generation gap when renewable investment is delayed. The Midwest combines both features and accounts for more than half of the avoided local air-pollution mortality damages. Electricity trade creates spatial spillovers: renewable investment in one region can reduce fossil generation and environmental damages elsewhere. California, for example, reduces gas generation partly by importing additional renewable electricity from other regions in the Western Interconnection.

Perfect-foresight counterfactuals separate uncertainty about future policy dates from the effects of changes in subsidy availability themselves. When future subsidy changes are known in advance, firms respond by shifting renewable investment across eligibility windows. Foresight therefore allows firms to adjust the timing of investment, but it does not eliminate the consequences of subsidy lapses. When interruptions in support are sufficiently long, renewable investment remains lower during the lapse, while fossil generation fills the gap. Continuous support can therefore remain valuable even when firms know the future policy path.

More broadly, the results show that policy stability matters during the green energy transition, even holding expected government spending fixed. The transition requires large, irreversible investments, while the policies supporting those investments can change over time. When investment is irreversible and costly to shift across time, policy changes affect not only how much firms invest, but when they invest, which technologies are built, and how the existing capital stock is used in the meantime. These effects can persist after support returns and, through trade, spill across regions. The design of transition policy therefore involves not only how much support governments provide, but also how reliably that support is provided over time.

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A Additional model details

A.1 Subsidy eligibility

The wind and solar eligibility indicators are

$$\chi_t^W = \mathbf{1}[s_t = 1], \qquad \chi_t^S = \mathbf{1}[t < 2028] + \mathbf{1}[t \geq 2028]\mathbf{1}[s_t = 1]. \tag{A.1}$$

Wind invested while the subsidy is active enters the eligible vintage $K_{jt}^{W,PTC}$ according to (4.4). The production tax credit is grandfathered. Solar receives the investment tax credit at construction.

A.2 Electricity prices and profits

Substituting the Pareto survival function (4.9) into the expected-price expression (4.11) gives the wholesale price in closed form. Define

$$x_{jt}^{d2} = Q_{jt}^0, \qquad x_{jt}^{d3} = Q_{jt}^0 + K_{jt}^{\text{lo}}, \qquad \bar{Q}_{jt} = Q_{jt}^0 + K_{jt}^{\text{lo}} + K_{jt}^{\text{hi}}.$$

Then

$$\begin{aligned} \bar{p}_{jt}^E = {} & mc_{jt}^{\text{lo}} \left(\frac{x_{jt}^{\min}}{x_{jt}^{d2}} \right)^{\theta_j} \\ & + (mc_{jt}^{\text{hi}} - mc_{jt}^{\text{lo}}) \left(\frac{x_{jt}^{\min}}{x_{jt}^{d3}} \right)^{\theta_j} \\ & + (V_j - mc_{jt}^{\text{hi}}) \left(\frac{x_{jt}^{\min}}{\bar{Q}_{jt}} \right)^{\theta_j}, \end{aligned} \tag{A.2}$$

with each term capped at its coefficient when the corresponding threshold lies below $x_{jt}^{\min}$. Additional zero-marginal-cost capacity raises the dispatch thresholds and lowers the expected wholesale price.

Regional net exports are

$$X_{jt} = G_{jt} - \bar{E}_{jt}.$$

The local price is related to the interconnection hub price by (4.12). The associated trans-

mission cost,

$$c_j(X_{jt}) = \frac{\gamma}{2}\frac{X_{jt}^2}{\overline{TL}_j},$$

is a real resource cost. The congestion basis itself is a transfer and is returned to households through $\mathcal{R}_{jt}$.

Electricity-sector profit is

$$\begin{aligned}\Pi_{jt}^E = p_{jt}^E \left[(CF_j^W K_{jt}^W + CF_j^S K_{jt}^S + Q_{jt}^{\text{base}})8760 + gen_{jt}^G + gen_{jt}^C\right] \\ + \phi^{PTC} CF_j^W 8760 K_{jt}^{W,PTC} - mc_t^G gen_{jt}^G - mc_t^C gen_{jt}^C.\end{aligned} \tag{A.3}$$

The first line is electricity-market revenue; the second adds the production tax credit earned by eligible wind and subtracts fossil fuel costs.

A.3 Capacity adequacy

I impose a capacity-adequacy requirement to ensure that each region maintains enough reliable generation capacity to meet forecast demand. Not all installed capacity contributes equally to system reliability. Gas, coal, nuclear, and hydro count fully toward the requirement, while wind and solar count according to their effective load-carrying capability. Accredited capacity is therefore

$$K_{jt}^{acc} = K_{jt}^G + K_{jt}^C + Q_{jt}^{base} + \zeta_j^W K_{jt}^W + \zeta_j^S K_{jt}^S, \tag{A.4}$$

where ζ_j^W and ζ_j^S measure the contribution of wind and solar capacity to system reliability. For example, $\zeta_j^W = 0.15$ means that one MW of installed wind capacity contributes 0.15 MW toward the capacity-adequacy requirement.

The system must satisfy

$$\Pr\left(D_{jt}^{fc} > K_{jt}^{acc}\right) \leq \varepsilon, \tag{A.5}$$

where D_{jt}^{fc} is exogenous forecast load and ε is the target probability that forecast load exceeds accredited capacity. Because the requirement is based on exogenous forecast demand, it does not respond to the model's endogenous electricity price or electricity demand.

Under the Pareto load distribution, (A.5) implies a minimum level of accredited capacity in each region. Because gas is the only dispatchable generation technology that can be built endogenously, gas investment fills any shortfall when existing accredited capacity falls below this requirement.

The multiplier on the capacity-adequacy constraint is the capacity price p_{jt}^{cap}. When the constraint binds,

$$p_{jt}^{cap} = \max\left\{p_t^G(r_t^K + \delta^G) - \mathrm{MPK}_{jt}^{G,\mathrm{energy}},\, 0\right\}. \tag{A.6}$$

The capacity price is the shadow value of additional reliable generation capacity. It equals the difference between the user cost of gas capacity and the return that gas earns in the energy market—the "missing money" required to support enough gas investment to satisfy the capacity-adequacy constraint. Because p_{jt}^{cap} is a shadow value rather than a resource cost, it does not enter aggregate welfare. Each region's capacity-adequacy requirement is determined by its own forecast demand.

A.4 Formal equilibrium definition

Within-period equilibrium. Given generation-capital stocks $\mathbf{K}_{jt}$, policy state s_t, and the exogenous demand and cost paths, a within-period equilibrium consists of regional electricity prices $\{p_{jt}^E\}$, interconnection hub prices $\{\bar{p}_{mt}^E\}$, generation and dispatch, household and firm electricity demands, and net trade such that:

1. each region dispatches generation in merit order and the wholesale price satisfies (4.11);
2. household and firm electricity demands satisfy (4.10);
3. local and hub prices satisfy (4.12);
4. electricity markets clear within each interconnection; and
5. congestion-basis payments are returned through $\mathcal{R}_{jt}$.

Dynamic equilibrium. A dynamic equilibrium is a within-period equilibrium in every period together with capital stocks, investment policies, financial wealth, consumption, and the national capital rental such that:

1. generation investment solves (4.5) subject to irreversibility, capital accumulation (4.3), and the wind subsidy vintage (4.4);
2. capital evolves according to its laws of motion;
3. the stochastic discount factor is consistent with household consumption;
4. the national production-capital market satisfies (4.8);
5. regional labor markets and the aggregate goods market clear;
6. the federal government budget balances;
7. the policy state follows Π; and
8. the economy converges to the terminal stochastic steady state.

The federal government finances subsidy outlays through the labor-income tax and national lump-sum transfers,

$$\sum_j \varsigma_t w_{jt} L_{jt} = \Phi_t + \sum_j N_j T_t^{mat}, \tag{A.7}$$

where Φ_t denotes aggregate subsidy expenditure.

A.5 Resource accounting

By aggregate goods-market clearing, the consumption component of welfare in (4.14) can equivalently be written

$$\begin{aligned} W^C = \sum_t \beta^t \Bigg[& \sum_j Y_{jt} - \sum_{j,k} p_t^k I_{jt}^k - \sum_{j,k} \Psi^k(I_{jt}^k, K_{jt}^k) \\ & - \sum_j \left(mc_t^G gen_{jt}^G + mc_t^C gen_{jt}^C \right) - \sum_j c_j(X_{jt}) - \Delta K_{P,t}^{\text{total}} \Bigg]. \end{aligned} \tag{A.8}$$

Generation investment, adjustment costs, fossil fuel use, transmission, and production-capital investment are real resource costs. Federal subsidies, the taxes that finance them, household electricity payments, and congestion-basis rebates are transfers and therefore cancel in aggregate welfare. Generation investment is valued at its full resource price p_t^k, rather

than the ITC-reduced private price $\tilde{p}_t^k$, so that the ITC remains a transfer rather than appearing as a reduction in resource cost.

B Computational algorithm

The model is a non-stationary stochastic spatial general equilibrium. I solve it by nesting the regional dynamic investment problems inside the within-period electricity-market and general-equilibrium fixed points.

B.1 Terminal stochastic steady state

Backward induction requires a terminal condition. I solve the stationary value function at terminal fundamentals by value-function iteration on a three-dimensional capital grid (K^S, K^W, K^G) crossed with the two policy states. At each grid point, the three investment choices $(I^S, I^W, I^G) \geq 0$ are optimized by coordinate-ascent golden-section search.

The eligible wind vintage $K^{W,PTC}$ is not an additional grid dimension. Conditional on eligibility, the production credit is separable from the energy value of wind, so its contribution can be represented by the present value

$$q^{W,PTC} = \frac{\phi^{PTC} CF_j^W \,\mathrm{HRS}}{1 - \beta(1-\delta^W)(1-h^{exit})}. \tag{B.1}$$

This value enters the marginal return to wind investment only when new capacity qualifies for the subsidy.

B.2 Backward-induction transition

Given the terminal value function, I solve the finite-horizon transition from 2010 to 2130 backward with time-varying fundamentals; quantitative results are reported through 2070.

B.3 Within-period and general-equilibrium fixed points

Within each period, a low-dimensional fixed point jointly determines price-elastic electricity demand, merit-order prices, and the scale of the Pareto load distribution. Interconnection hub prices clear net trade among member regions. An outer fixed point jointly determines state-contingent local electricity prices, the national capital rental, and aggregate consumption.

C Illustration of the investment mechanism

I solve the full model for the Eastern Interconnection along a deterministic policy path in which the subsidy is active, lapses for six years, and is then permanently restored. Figure 11 follows the Midwest, where federal support is important for marginal wind investment.

While the subsidy is active, Midwest wind investment runs at approximately 11 GW per year. When the subsidy lapses, newly built wind no longer qualifies for the PTC and wind investment falls to zero.

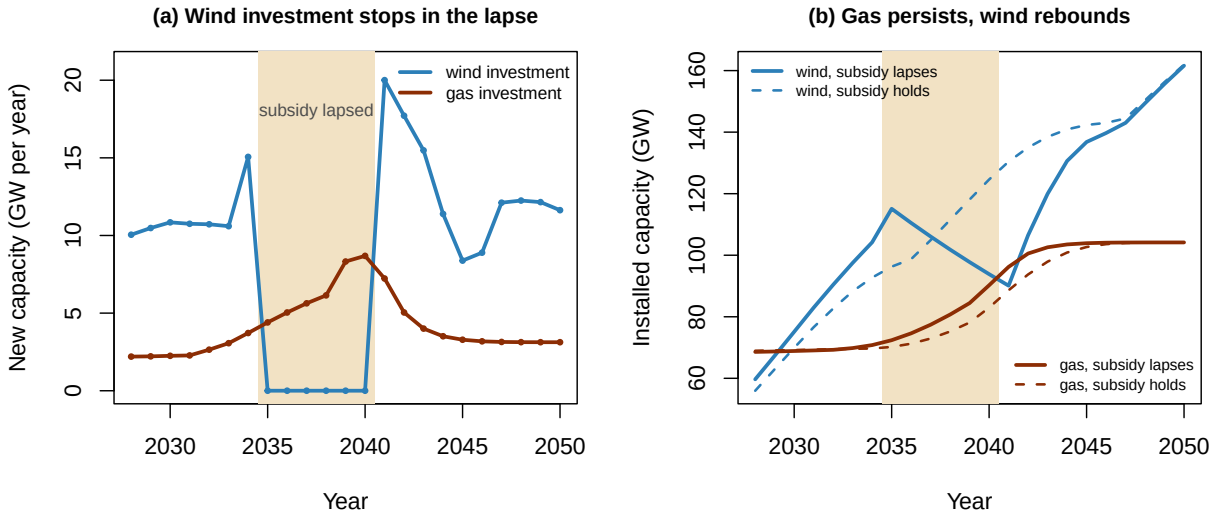


Figure 11: Investment and capacity around a temporary subsidy lapse.
Notes. Full model solved for the Eastern Interconnection along a deterministic policy path in which the subsidy is active, lapses over 2035–2040, and is then restored (the Midwest).

D Policy-process calibration

This appendix reports the pre-2028 policy beliefs and the political transition matrix used to construct the policy process in Section 5.3.

D.1 Pre-2028 policy path

Table 8: Common pre-2028 policy path.

Years	Belief	Pr(lapse | active)	Event
2010	active	—	initial state $s_0 = 1$
2010–2021, off-cliff	active, low lapse risk	0.08	
2012, 2014, 2016, 2019, 2021	scheduled expiration	0.60	PTC expiration dates
2022–2027	active	0.00	IRA commitment
2027	lapse in 2028	1.00	OBBBA repeal
2028 onward	regime-specific	—	counterfactual branch

Notes. The pre-2028 path is common to every counterfactual. The counterfactual policy processes begin in 2028.

D.2 Federal political control

I group each Congress from 1993 through 2025 as a Democratic unified control, Republican unified control, or divided government. Table 9 reports the transition frequencies.

Table 9: Federal political-control transitions, 1993–2025.

	Next Congress		
Current Congress	Democratic trifecta	Republican trifecta	Divided
Democratic trifecta	0.00	0.00	1.00
Republican trifecta	0.00	0.33	0.67
Divided	0.20	0.30	0.50

Notes. Transition frequencies are calculated across two-year Congresses. The ergodic frequencies of Democratic trifecta, Republican trifecta, and divided government are 0.12, 0.27, and 0.61, respectively.

E Calibration and data

Table 10 reports regional summary statistics, and Figure 12 plots the principal exogenous paths.

Table 10: Summary statistics: the U.S. wind, solar, and gas sector by region.

Region	Inter-connect	Wind CF	Solar CF	Wind (GW)	Solar (GW)	Gas (GW)	Demand (TWh)
Southwest	West	0.38	0.32	5.7	7.6	20.7	117
Central	East	0.37	0.26	28.8	0.6	24.8	263
Midwest	East	0.34	0.21	41.3	15.5	98.0	667
Texas	ERCOT	0.34	0.29	42.3	22.5	64.9	403
Northwest	West	0.34	0.29	20.0	12.0	29.9	347
Carolinas	East	0.28	0.25	0.4	8.4	25.6	214
California	West	0.27	0.31	6.5	22.3	34.6	249
Mid-Atlantic	East	0.27	0.19	3.7	11.2	90.3	751
New England	East	0.26	0.19	1.6	3.1	15.8	112
New York	East	0.23	0.19	2.9	2.7	20.7	140
Tennessee	East	0.10	0.25	0.0	0.6	6.9	186
Southeast	East	0.04	0.27	0.0	5.7	35.3	227
Florida	East	0.04	0.27	0.0	10.9	54.6	241
U.S. total				153	123	522	3,919

Notes. Capacity factors are net generation over nameplate capacity, 2019–2024 average (EIA-923/EIA-860); regions with negligible wind show 0. Installed capacity is 2024 (EIA-860). Demand is 2024 electricity sales (AEO 2025). Sources: EIA-860, EIA-923, EIA AEO 2025.

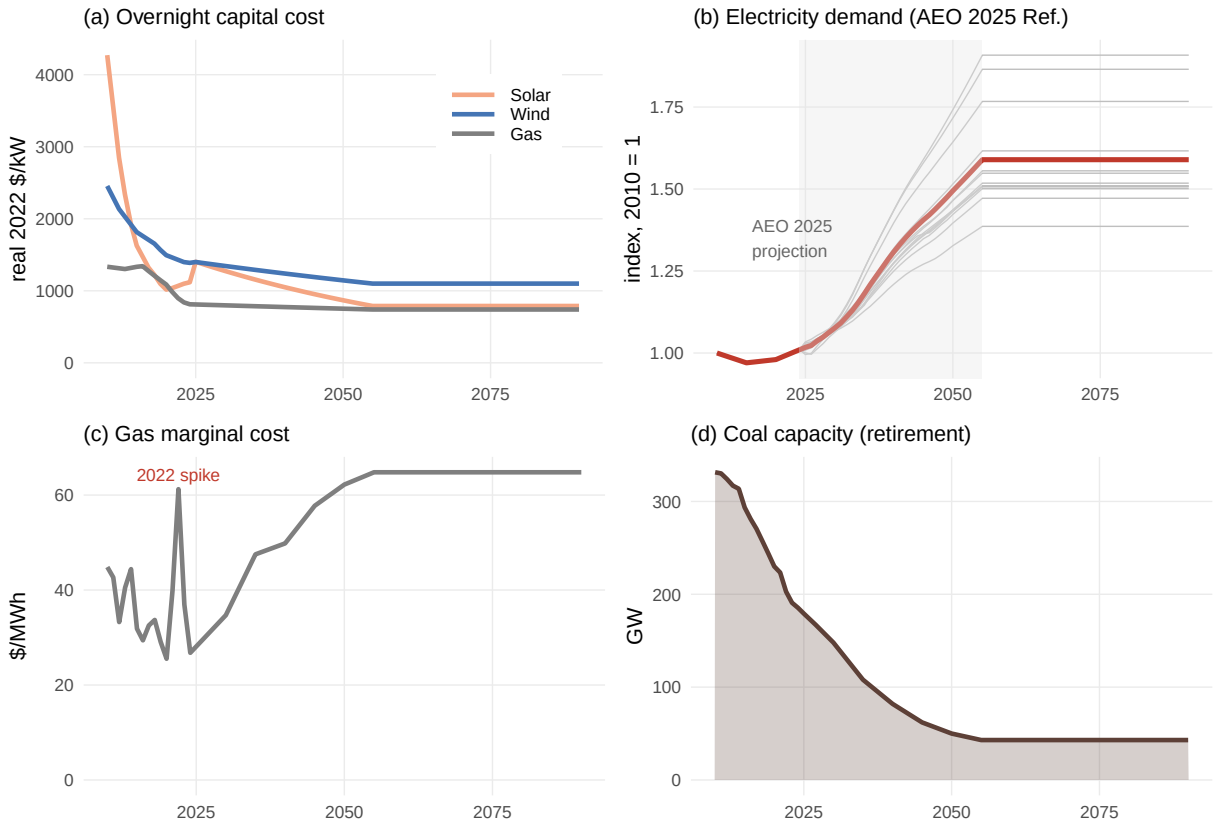


Figure 12: The exogenous processes that drive the transition.

Notes. Panel (a): overnight capital costs (real 2022 $/kW). Panel (b): electricity-demand index (2010=1). Panel (c): gas marginal cost ($/MWh). Panel (d): coal capacity (GW). Sources: NREL ATB, LBNL/EIA, and EIA AEO 2025.

E.1 Capacity, generation, and capacity factors

Initial generation stocks and the historical coal-retirement path come from EIA-860 and are aggregated from generators to the 13 model regions by technology. After the observed period, national coal retirements follow the EIA AEO path and are allocated across regions.

Regional capacity factors are calculated as the ratio of EIA-923 net generation by EIA-860 nameplate capacity (averaged over 2019–2024):

$$CF_j^k = \frac{\text{net generation}}{\text{nameplate capacity} \times 8760}.$$

Gas is represented using the combined-cycle capacity factor. Fossil marginal costs combine delivered fuel prices with heat rates and transition toward the AEO 2025 path after the historical period.

E.2 Demand and hourly load

Regional annual electricity demand follows the EIA AEO 2025 Reference case, cross-walked to the 13 model regions. For each region, I estimate the Pareto tail parameter θ_j by maximum likelihood using the upper tail of hourly demand (EIA-930).

E.3 Prices, trade, and emissions

The regional trade slope is

$$s_j = \frac{\gamma}{\overline{TL}_j}.$$

For regions with observed hub prices, I estimate the slope using the average wholesale-price basis and average net position. For the remaining regions, I impute it from a common congestion parameter γ and observed gross interchange capacity. Net positions and interchange are from EIA-930. Texas is treated as a separate interconnection with no trade with the other model regions.

Emissions factors are $\xi_G = 0.40$ and $\xi_C = 0.95$ tCO_2/MWh from EPA eGRID.

E.4 Federal tax-credit history

Figure 13 summarizes the statutory history used to construct the historical policy path.

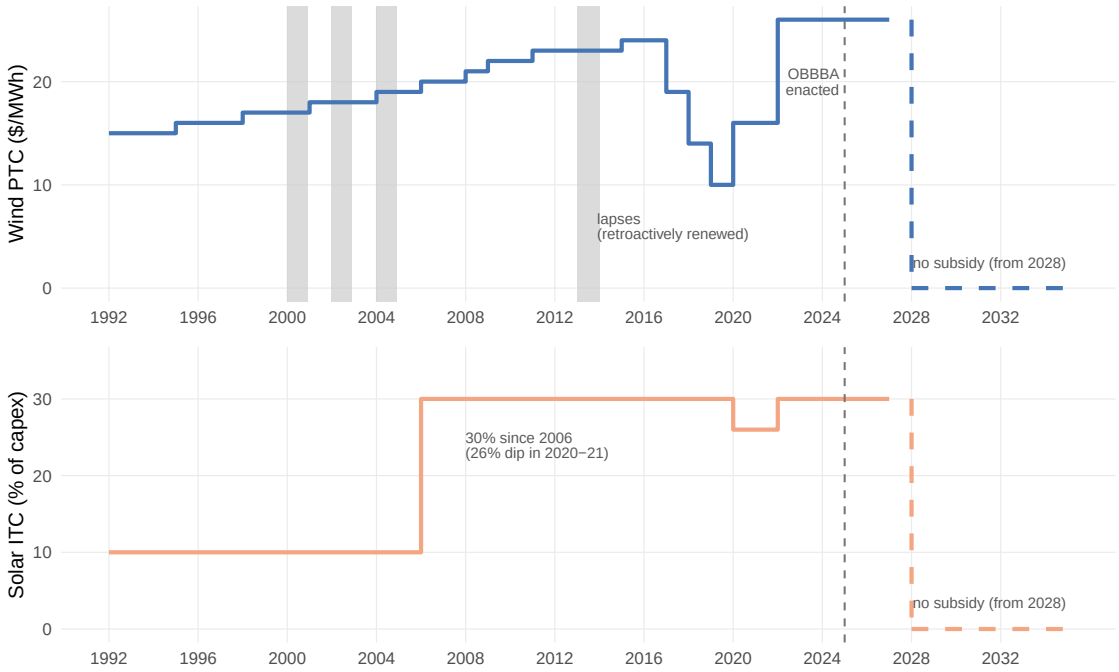


Figure 13: Federal wind and solar tax-credit history and scheduled termination, 1992–2035.

Notes. Source: enacting statutes and IRS notices. The wind PTC and solar ITC remain available for projects placed in service through 2027 under the 2025 One Big Beautiful Bill Act; the dashed segments show the scheduled termination from 2028. Projects beginning construction by mid-2026 may remain eligible if placed in service by 2030 under safe-harbor rules.

Table 11: The production tax credit in the economics of a representative wind project.

	$/MWh
Revenue per MWh, during the ten-year credit period	
Wholesale energy price	50
Production tax credit (§45)	26
Total	76
Levelized cost per MWh, unsubsidized	
Capital ($1,600/kW, 25 yr, 5%)	39
Fixed O&M ($40/kW-yr)	14
Levelized cost of energy	53
Net profit per MWh	
Without the credit	−3
With the credit	+23
Credit as a share of revenue	34%

Notes. All entries are dollars per megawatt-hour generated.

Table 12: Interconnection trade slope by region.

Region	Interconnection	X_j (GW)	$\overline{TL}_j$ (GW)	s_j	Source
Mid-Atlantic	Eastern	+5.9	7.9	0.24	direct
Central	Eastern	+1.1	1.1	1.79	imputed
Southeast	Eastern	+1.0	2.0	1.00	imputed
New England	Eastern	-0.2	0.2	8.56	imputed
Carolinas	Eastern	-0.3	1.1	1.80	imputed
Tennessee	Eastern	-0.4	3.0	0.68	imputed
Florida	Eastern	-0.4	0.4	4.98	imputed
New York	Eastern	-1.6	2.1	0.99	imputed
Midwest	Eastern	-2.7	6.0	3.58	direct
Northwest	Western	+3.5	3.9	0.52	direct
Southwest	Western	+2.1	4.0	0.52	imputed
California	Western	-6.0	6.0	0.34	imputed
Texas	Texas	-0.2	0.2	12.80	autarky

Notes. The trade slope $s_j = \gamma/\overline{TL}_j$ is the wholesale price basis a region pays per GW of net exports X_j (positive = net exporter). For regions with hub prices (ICE, and NYISO for New York), s_j is estimated *directly* as the mean price basis over the mean net position; for the others it is *imputed* as $\gamma/\overline{TL}_j$ from a common congestion constant $\gamma \approx 2$ (the median of $s_j\overline{TL}_j$ across the priced regions) and the gross-interchange proxy $\overline{TL}_j$. Sources: ICE, NYISO, EIA-930.

F Additional model validation

Section 5.4 reports the principal untargeted moments. This appendix provides the corresponding dynamic and regional comparisons.

F.1 Capacity

Figure 14 compares modeled and observed national wind and solar capacity over 2010–2024 and the 2024 regional cross-section. National wind and solar capacity levels enter the internal calibration; their allocation across regions does not. The model reproduces the regional ordering of wind capacity particularly closely, with a Spearman rank correlation of 0.96, while the corresponding solar correlation is 0.57.

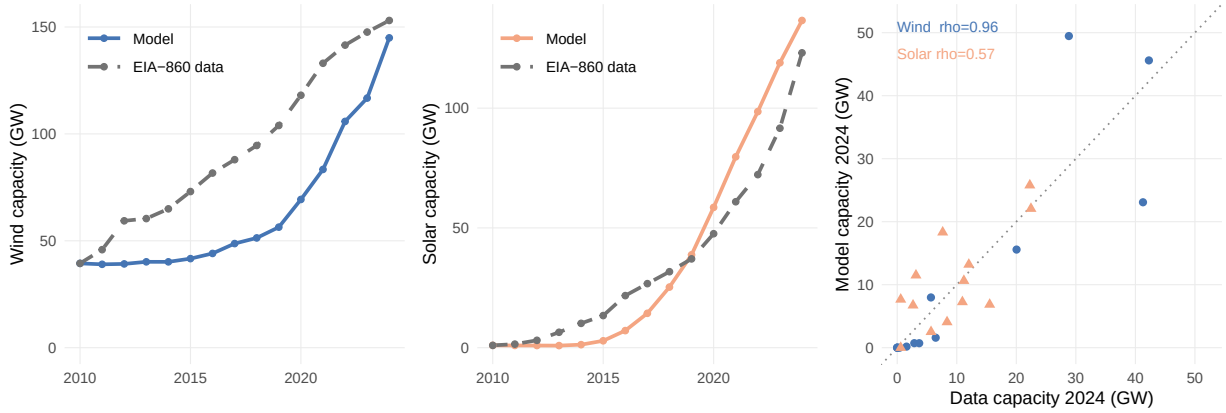


Figure 14: Model versus data: capacity, 2010–2024.

Notes. Panels (a)–(b) compare national wind and solar capacity in the model and EIA-860 data. Panel (c) compares modeled and observed 2024 capacity across the 13 regions. Source: EIA-860.

F.2 Wholesale electricity prices

Figure 15 compares modeled wholesale prices with observed trading-hub prices by interconnection. The model reproduces the broad price level across interconnections and ERCOT as the low-price region.

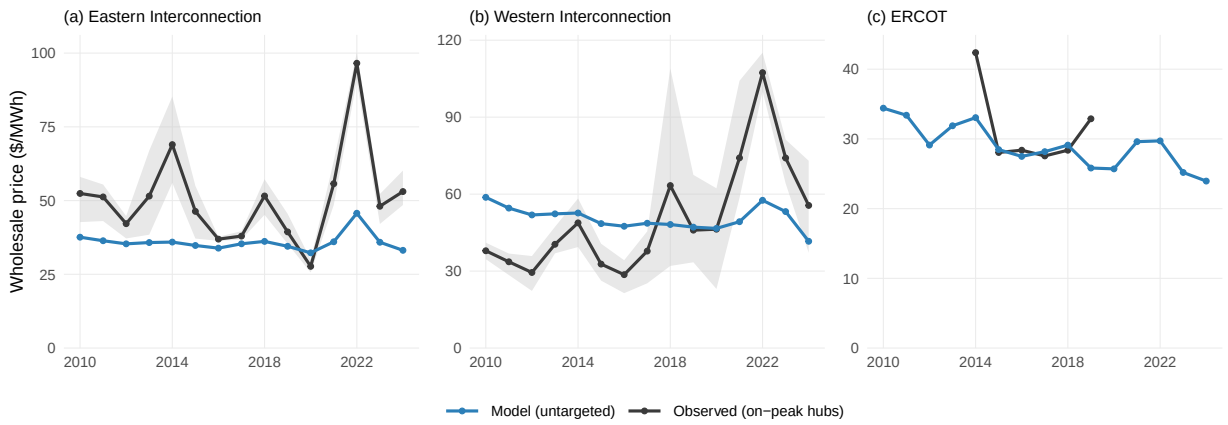


Figure 15: Untargeted wholesale price, by interconnection.

Notes. The black line is the mean across hubs and the grey band is the range. Eastern hubs are PJM West, the Indiana Hub, and the Mass Hub; western hubs are Mid-C, NP15, SP15, and Palo Verde. Source: EIA wholesale market data (ICE hub prices).

F.3 Local air-pollution damages

Source damages. I use AP4, the fourth installment of the Air Pollution Emission Experiments and Policy model (Dennin et al., 2024), with a 2017 emissions and population base. AP4 treats each electric generating unit as a point source with its own effective-stack-height source–receptor matrix and computes the marginal mortality damage, in dollars per metric ton, of its NH_3, NO_x, primary $PM_{2.5}$, SO_2, and VOC emissions through their contribution to ambient fine-particulate concentrations. The endpoint is $PM_{2.5}$-related mortality, valued at the value of a statistical life; "$PM_{2.5}$-related mortality" refers to this health endpoint, mortality from ambient fine-particulate exposure, and not only to direct emissions of $PM_{2.5}$. A source damage is the total contiguous-U.S. mortality damage from one additional metric ton emitted at that source. AP4 monetary outputs are in real 2020 dollars.

Matching to generation. AP4 generating units are identified by ORIS plant code. I match them to EIA-923 net generation (U.S. Energy Information Administration, 2017) by ORIS, aggregating AP4 point sources to the ORIS plant level. I classify plants as coal or natural gas using EIA fuel and technology information.

Plant-level damages. For plant i and pollutant p, let MD_{ip} be the AP4 marginal damage (dollars per metric ton), E_{ip} observed annual emissions (metric tons), and G_i net generation

(MWh). The plant damage rate, in dollars per MWh, is

$$d_i = \frac{\sum_p MD_{ip}\, E_{ip}}{G_i}. \tag{F.1}$$

This combines location-specific damages per ton, observed emissions intensity, and observed generation.

Region–fuel coefficients. I assign each plant to one of the thirteen model regions. For region j and fuel $k \in \{\text{coal}, \text{gas}\}$, the generation-weighted damage coefficient is

$$d_{jk} = \frac{\sum_{i \in (j,k)} G_i\, d_i}{\sum_{i \in (j,k)} G_i}. \tag{F.2}$$

The national generation-weighted rates are \$105/MWh for coal, \$11/MWh for existing gas, and \$4/MWh for recently built combined-cycle gas (AP4 2020 dollars). Table 13 reports the region-by-fuel coefficients.

Table 13: AP4 region-by-fuel local-pollution damage coefficients.

	Damage (\$/MWh, 2020\$)		Plants	
Region	Coal	Gas	Coal	Gas
CAL	—	5.8	0	60
CAR	74.3	6.2	14	26
CENT	96.2	3.6	21	69
FLA	84.9	10.4	7	51
MIDA	136.6	21.0	56	152
MIDW	123.5	12.8	98	227
NE	90.4	3.7	2	50
NW	32.4	3.7	30	67
NY	983.3†	10.1	3	66
SE	40.5	8.0	9	45
SW	41.4	6.2	8	34
TEN	109.1	2.8	6	4
TEX	126.4	9.9	19	122
National	105.3	11.0	273	973

Notes. Generation-weighted AP4 marginal damage per MWh by region and fuel, in 2020 dollars, before conversion to 2022 dollars. National rates are generation-weighted across all matched plants. †The New York coal coefficient is based on only three plants generating 0.9 TWh.

Application. A positive value is a damage avoided under continuous support, since the uncertain policy generates more fossil power. The coefficients are converted to 2022 dollars (Appendix F.4) and the annual stream is discounted at the model's 4% rate to 2010, giving $46 billion.

Table 14: Avoided local air-pollution mortality damages by source region.

Source region	Coal	Gas	Total
Midwest	23.7	1.3	24.9
Texas	11.8	0.4	12.2
Mid-Atlantic	0.0	2.5	2.5
Northwest	1.3	0.1	1.4
California	0.0	1.1	1.1
Central	0.8	0.1	0.9
Southwest	0.7	0.1	0.8
Carolinas	0.8	0.0	0.8
Florida	0.1	0.6	0.7
New York	0.0	0.4	0.4
New England	0.0	0.2	0.2
Southeast	0.1	0.1	0.1
Tennessee	0.0	0.0	0.0
National	**39.2**	**6.9**	**46.1**

Notes. Present value of avoided $PM_{2.5}$-related mortality damages (permanent minus budget-neutral uncertain), in billions of real 2022 dollars, discounted at 4% to 2010, by the source region of the emitting fossil plants. Damages are attributed to the region containing the emitting plants, not to the region whose residents are exposed; AP4 integrates exposure across the contiguous United States.

F.4 Welfare valuation, dollar years, and discounting

All monetary welfare values in the paper are expressed in real 2022 dollars, as a present value to 2010, over a horizon through 2070. The present-value base year of 2010 is the model's native reporting base; because the counterfactuals share the realized policy history through 2027, substantive differences between them begin only after the 2028 policy branch, up to numerical error in the stochastic baseline.

Consumption and resource cost. The model's real costs are calibrated in 2022 dollars. Consumption is discounted at the household rate implied by $\beta = 1/1.04$, so $PV_C = \sum_t \Delta C_t/(1.04)^{t-2010} = \$113.5$ billion.

Climate damages. I value avoided CO_2 with the EPA emissions-year-specific social cost of carbon under the 2% specification (U.S. Environmental Protection Agency, 2023). The EPA series is published in 2020 dollars; I convert the entire series once to 2022 dollars and do not inflate it forward year by year, because the EPA values are already emissions-year specific in constant dollars. With SCC_t the converted value and E_t emissions, $\text{ClimateBenefit}_t = (E_t^U - E_t^P)\, SCC_t$ and $PV_{climate} = \sum_t \text{ClimateBenefit}_t/(1.02)^{t-2010} = \$359.6$ billion.

Local air pollution. AP4 damages are in 2020 dollars, converted once to 2022 dollars. Unlike the social cost of carbon, AP4 reports contemporaneous monetized mortality damages from a given year's emissions and does not embed a future stream of damages. I discount the annual local-pollution stream at the model's private 4% rate as a modeling convention for contemporaneous monetized damages, giving $PV_{local} = \$46.1$ billion.

Inflation. External monetary estimates are converted to 2022 dollars with the BEA GDP implicit price deflator (U.S. Bureau of Economic Analysis, 2024): the annual index rises from 105.4 in 2020 to 118.0 in 2022, a factor of 1.120. Changing the dollar year is not an economic change; it places consumption, climate damages, and local-pollution damages in common units.

Discount rates. The components use different rates by design: consumption at the private rate implied by $\beta = 1/1.04$; climate damages are valued using the EPA's year-specific social cost of carbon under its 2% discount-rate scenario; and local mortality damages at the private rate. I do not impose a single social discount rate.

Table 15: Welfare valuation conventions.

Component	Dollar year	PV year	Discount rate
Consumption / resource cost	2022$	2010	4%
CO_2 damages	2022$	2010	2% (EPA)
Local mortality damages	2022$	2010	4%

Table 16: Value of continuous support across forward regimes (present value, SCC = \$213/tCO$_2$ in 2022 dollars).

Permanent subsidy vs.	Climate benefit (\$B)	Consumption cost (\$B, PV)	Local air pollution (\$B)	Value (compreh.) (\$B)	Excl. local air poll. (\$B)	Breakeven SCC (\$/tCO$_2$)
Budget-neutral uncertain subsidy	360	113	46	+292	+246	67
No subsidy	563	247	70	+385	+315	94
Cross-party	53	5	8	+56	+48	21

Notes. As Table 5, for the forward regimes. The comprehensive value adds avoided local air-pollution ($PM_{2.5}$) mortality damages (AP4, discounted at 4%) to the climate-only net value; the *Excl. local air poll.* column reports the climate-only net value. The breakeven social cost of carbon in the last column is computed on the climate channel. *No subsidy* is the permanent repeal. The value is positive and the breakeven lies far below the EPA's \$213 in every regime.

G Robustness

The value of continuous subsidy reported in this appendix is comprehensive, including both avoided climate damages and avoided local air-pollution mortality damages; the tables also report the climate-only value, excluding local air pollution.

G.1 Policy persistence

The expected duration of a lapse is the main uncertainty in the policy calibration. I compare the baseline with a faster renewal regime, in which the subsidy is active 85% of the time in the long run, and with permanent repeal. Table 16 reports the results.

G.2 Climate discount rate

In the baseline, private consumption is discounted at the model's market rate while climate damages are discounted at the social rate used in the EPA SCC calculation. Table 17 varies the rate applied to climate damages. Even when climate damages are discounted at the market rate, the value of continuous subsidy remains positive.

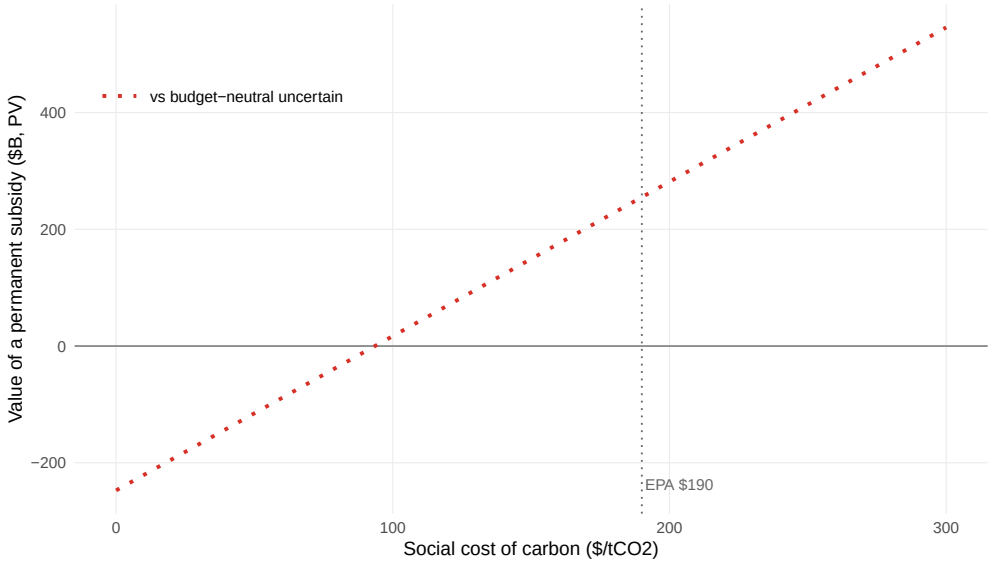


Figure 16: Value of continuous subsidy and the social cost of carbon.
Notes. Welfare difference between the permanent and budget-neutral uncertain subsidies as the social cost of carbon varies.

Table 17: Welfare results across the climate discount rate.

	Value of continuous support (permanent − uncertain)			Value of the subsidy (uncertain − no subsidy)		
Climate rate	Climate ($B)	Value ($B)	Breakeven SCC	Climate ($B)	Value ($B)	Breakeven SCC
1.5%	433	+320	56	247	+113	115
2.0% (baseline)	360	+246	67	203	+69	140
3.0%	251	+137	96	139	+5	206
4.5% (market)	150	+37	161	81	−53	354

Notes. The value of continuous support over the budget-neutral uncertain subsidy (columns 1–3) and of the uncertain subsidy over no subsidy (columns 4–6), as the rate used to discount the climate benefit varies. Only the climate benefit is re-discounted; the consumption cost is held at its present value at the model rate ($113 billion for continuous support, $134 billion for the subsidy), since the discount factor β is a structural parameter. The baseline is the EPA's 2% social rate; the 4.5% row discounts the climate benefit at the model market rate, so the climate externality is treated like private consumption. Value = climate − consumption cost; breakeven SCC is the social cost of carbon at which the value is zero (the reference path is scaled from the EPA central value, $\$213/tCO_2$ in 2022 dollars). The value of continuous support stays positive at every rate; the value of the subsidy turns negative once the climate benefit is discounted at the market rate.

G.3 Perfect-foresight investment by technology

Figure 17 separates the renewable-investment response in Section 7.3 into wind and solar. The sharp anticipatory response is concentrated in wind, for which eligibility determines the stream of production-tax-credit payments associated with newly qualifying capacity.

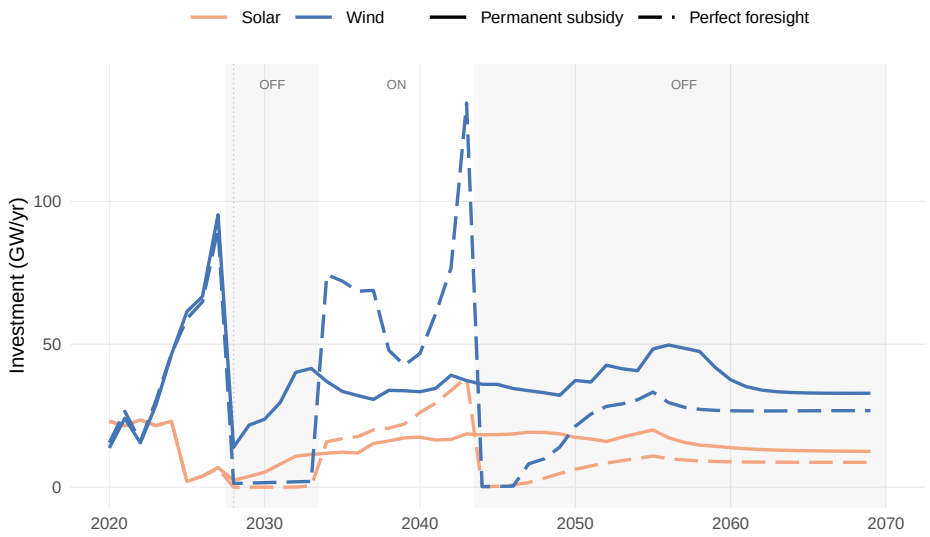


Figure 17: Wind and solar investment under perfect foresight.

Notes. Annual wind and solar investment under permanent support and the deterministic perfect-foresight policy. Shaded areas indicate periods in which the subsidy is unavailable under the foresight path.

H Additional counterfactual results

H.1 Regional results

Figure 18 reports the permanent-minus-uncertain capacity gaps and uncertain-minus-permanent emissions gap for all 13 regions. The largest wind-capacity differences occur in the interior regions where wind resources are strong and federal support affects the marginal investment decision. The gas-capacity and emissions responses need not occur in the same regions as the renewable-investment response because electricity is traded within interconnections.

Table 18 reports the regional decomposition of the value of the uncertain subsidy relative to no subsidy.

Table 18: The value of the uncertain subsidy by region, decomposed (uncertain versus no subsidy).

	Climate gains				Local poll.†	Consumption					
Region	Coal	Ex. gas	New gas	Total	Avoided	Invest.	Income	Wages	Profit	Total	Value
Midwest	56	18	7	81	15	−45	29	0	0	−15	66
California	0	22	2	24	0	−16	6	2	−3	−11	13
Mid-Atlantic	0	15	3	18	2	−10	5	0	0	−6	12
Florida	2	6	0	7	0	−5	2	0	0	−3	4
New England	0	9	1	10	0	−9	4	1	−2	−6	4
Carolinas	5	0	0	5	1	−2	0	0	0	−2	3
New York	0	7	0	7	0	−6	3	1	−1	−4	3
Southeast	1	1	0	2	0	−1	0	0	0	−1	1
Texas	15	4	0	19	4	−49	34	−4	1	−18	1
Southwest	3	2	0	6	0	−6	−2	1	0	−7	−2
Tennessee	0	3	0	3	0	−11	1	0	0	−10	−7
Northwest	14	−2	1	14	1	−20	−8	7	−3	−24	−10
Central	0	2	5	6	0	−23	−7	3	0	−27	−21
National	**95**	**63**	**45**	**203**	**24**	**−203**	**66**	**12**	**−9**	**−134**	**69**

Notes. Present value in billions of dollars, by region, sorted by value. Climate gains value the region's own avoided CO_2 at the EPA year-specific social cost of carbon discounted at 2%, split by source (existing coal, existing gas, new gas). Consumption is the change in the present value of the region's consumption; income, wages, and profit raise it, generation investment enters negatively. Value = climate gains + consumption. Rows sum to the National row. †The final column reports avoided local air-pollution ($PM_{2.5}$) mortality damages attributed to emissions from fossil plants *located in* the region (AP4, discounted at 4%); it is shown for reference, is *not* added to Value, and is not a measure of health benefits received by residents of the region.

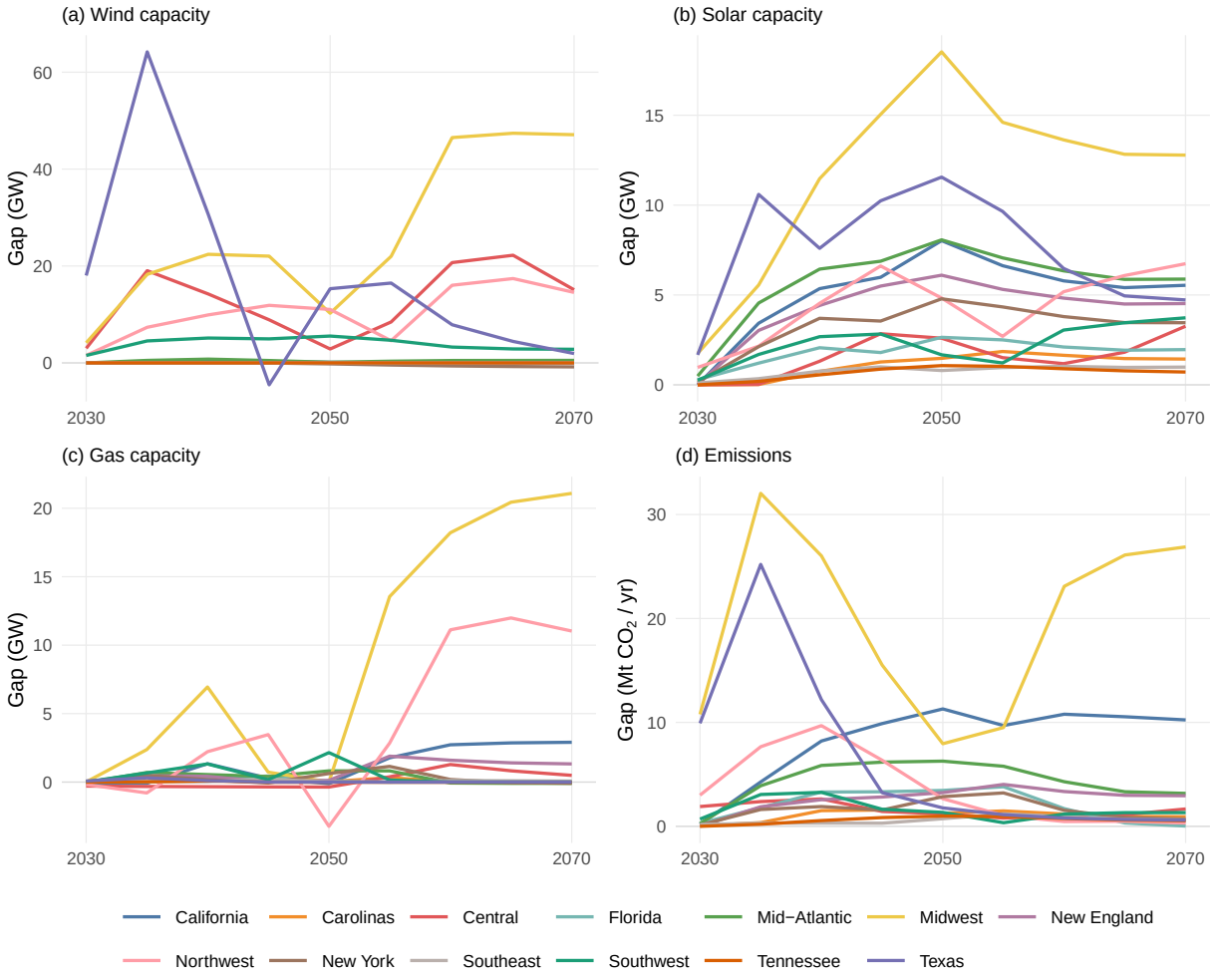


Figure 18: Regional gaps over time, all thirteen regions.

Notes. Gaps between a permanent subsidy and the budget-neutral uncertain subsidy at five-year intervals. Panels (a)–(b) report wind and solar capacity as permanent minus uncertain. Panels (c)–(d) report gas capacity and annual CO_2 emissions as uncertain minus permanent.